

Bottom-Up Mixed-Frequency Data Sampling (BUMIDAS)

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Abstract

We derive the bottom-up functional form of mixed-frequency data sampling (BUMIDAS) models for temporally aggregated VARMA processes. The required lag dimension depends on the disaggregated data-generating process and is invariant to aggregation, simplifying unrestricted estimation. BUMIDAS yields the same population forecast as recursive bottom-up forecasting when the relevant high-frequency information, including the target's own history, is observed or recoverable. A mean-squared forecast error decomposition separates information loss from forecast inefficiency. When the high-frequency target is proxied or unavailable, information criteria select finite approximations of a state-space filter. Simulations and applications to interest rates, exchange rates, inflation, and GDP nowcasts show substantial gains over restricted and unrestricted MIDAS, particularly for persistent processes under daily-to-monthly and daily-to-quarterly aggregation.

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1 Introduction

This study advances our understanding of how to construct direct mixed-frequency forecasts of temporally aggregated data, such as monthly averages of exchange rates or quarterly sums of output. Forecasting from temporally aggregated data can substantially reduce accuracy (Tiao, 1972; Amemiya and Wu, 1972; Kohn, 1982). The mixed data sampling (MIDAS) framework mitigates this loss by projecting the low-frequency target on high-frequency predictors (Ghysels et al., 2004). In the standard setting, the target’s own high-frequency observations are unavailable. Restricted MIDAS (RMIDAS) uses parsimonious lag-weighting functions (Ghysels et al., 2007; Andreou et al., 2010, 2013), while unrestricted MIDAS (UMIDAS) estimates a separate coefficient for each included high-frequency lag (Forni et al., 2015a, 2019). However, for a given low-frequency lag window, the number of unrestricted coefficients grows with the degree of aggregation, increasing estimation uncertainty, while restricted weights may not capture the relevant dynamics. We examine which high-frequency information is required for efficient forecasting and how parsimoniously that information can be used.

Our first contribution is to derive the optimal bottom-up direct mixed-frequency representation for temporally aggregated VARMA processes. We obtain this representation by aggregating optimal high-frequency forecasts, rather than deriving the aggregate process through polynomial transformations (Brewer, 1973; Marcellino, 1999; Forni et al., 2015a). We show that the optimal direct forecast is obtained by projecting the low-frequency target on only the most recent p high-frequency observations and q innovations of the underlying VARMA(p, q) process, provided these are observed or recoverable. We call this representation bottom-up MIDAS (BUMIDAS). When the forecast target depends on its own past, efficiency requires using its recent high-frequency observations. The result also holds for point-in-time forecasts and, after accounting for the observed part of the aggregate, nowcasts. Temporal aggregation changes the coefficients, but not the number of

required high-frequency terms, permitting parsimonious unrestricted estimation even for daily-to-monthly and daily-to-quarterly forecasts. BUMIDAS provides a complete-information benchmark with the same population forecast as recursively forecasting and aggregating the high-frequency observations (Zellner and Montmarquette, 1971; Kohn, 1982; Lütkepohl, 1986).

Our second contribution is a mean-squared forecast error (MSFE) decomposition that separates irreducible forecast uncertainty, losses from missing high-frequency information, and inefficiencies in how the available information is used. Complementing analyses of how temporal aggregation changes mixed-frequency specifications (Foroni et al., 2019; Martins and Teles, 2026), the decomposition distinguishes improvements in the information set from improvements in the forecasting rule. High-frequency observations of other variables can reduce information loss without recovering the predictive content of the target’s own high-frequency history. Changing lag structures or estimation methods cannot eliminate the remaining information loss.

Our third contribution characterizes how proxies for the high-frequency target can close this information gap. Following the reduced-form logic of Bai et al. (2013), we use a state-space representation to trace how the implied projection weights change with proxy quality. An uninformative proxy receives zero weight, leaving the standard MIDAS information set of aggregate target observations and high-frequency predictors. A perfect proxy reveals the high-frequency target and recovers the BUMIDAS benchmark. Between these limits, the best feasible forecast may depend on earlier proxy observations, observed high-frequency predictors, and aggregate dynamics. BUMIDAS therefore complements existing MIDAS methods by linking feasible specifications to the available information.

In VAR simulations with all high-frequency variables observed, specification tests corroborate the parsimonious BUMIDAS structure, including the redundancy of additional high-frequency lags and low-frequency autoregressive and moving-average terms. Information criteria can recover this

structure without prior knowledge of the underlying lag orders. Comparisons of AIC, BIC, and HQIC yield practical guidance consistent with the VAR lag-selection literature: BIC generally recovers the parsimonious specification most reliably, while HQIC better retains weak cross-variable effects in quarterly samples. When the high-frequency target is proxied or unavailable, selected specifications combine available high-frequency observations and aggregate history to approximate the best forecast attainable from those data. They may retain only a few high-frequency terms, exclude weak proxies, and include more than p aggregate lags. These findings help explain the empirical success of smaller MIDAS specifications (Kronenberg et al., 2023; Benmoussa et al., 2026) and motivate exploration of shrinkage approaches (Mogliani and Simoni, 2021; Babii et al., 2022; Kohns and Potjagailo, 2025).

When the target is observed at high frequency, out-of-sample simulations show that BUMIDAS closely matches efficient recursive bottom-up forecasts, with most gains retained when lag orders are selected empirically. The BIC-selected implementation (BUMIDAS-BIC) can reduce MSFE by more than 40 percent relative to the low-frequency VAR. For persistent processes, the gains under daily-to-monthly and daily-to-quarterly aggregation are commensurate with theory (Tiao, 1972; Kohn, 1982; Lütkepohl, 1986) and can exceed those documented in earlier studies based on monthly or quarterly data (see, e.g., Zellner and Montmarquette, 1971; Lütkepohl, 1986; Athanassopoulos et al., 2011). At the monthly frequency, reductions relative to UMIDAS and RMIDAS are approximately 17–26 and 12–19 percent, respectively, with equal predictive accuracy rejected at the 10 percent level in most replications. Gains against UMIDAS are particularly large under quarterly aggregation. BUMIDAS-BIC can also deliver lower monthly and quarterly MSFE than UMIDAS and RMIDAS when the target is observed only at low frequency and predictors are available at high frequency. Conventional MIDAS methods remain competitive at the weekly frequency in incomplete-information settings.

Large-sample simulations isolate the value of the information set by quantifying forecast losses that changes in model specification alone cannot eliminate. MSFE exceeds the complete-information benchmark by approximately 20–38 percent with low-frequency data alone and 6–20 percent when the high-frequency data of the target is not observed. Informative proxies can substantially reduce the remaining loss: a strong proxy lowers monthly excess MSFE from 20 to 4.4 percent. At monthly and quarterly frequencies, even moderate and weak proxies can recover a larger share of the remaining information loss, but substantial weekly gains require a strong proxy. In finite samples, however, the estimation cost of a larger specification can offset the information gains from using proxies.

We demonstrate these gains in out-of-sample forecasts of U.S. interest rates, exchange rates, and headline inflation, and in real-time GDP nowcasts. For interest rates and exchange rates, BUMIDAS closely matches recursive bottom-up forecast accuracy. One-quarter-ahead interest-rate MSFE falls by approximately 59 percent relative to the period-average model. For monthly-average exchange rates, most gains come from conditioning on the latest daily rate rather than the monthly average; crude oil adds little predictive content. BUMIDAS-BIC performs particularly well in the short inflation sample, where excluding the daily CPI proxy from the Billion Prices Project (Cavallo and Rigobon, 2016) leaves its forecast accuracy unchanged, suggesting little incremental predictive value of the proxy. For GDP nowcasting, BUMIDAS-BIC has the lowest first-month MSFE, reducing it by approximately 21–60 percent relative to bridge and conventional MIDAS methods.

These results expand the mixed-frequency forecasting toolkit. They show when direct forecasts can match recursive bottom-up efficiency, extending the literature on direct versus recursive forecasting (Ing, 2003; Marcellino et al., 2006; Chevillon, 2007). The bottom-up approach guides the use of high-frequency information, showing that MIDAS can be simpler to implement and more efficient than previously understood.

2 Bottom-Up MIDAS

2.1 Bottom-Up MIDAS: A General Result

Let us start with the existence of a high-frequency (daily) VARMA(p,q), given by:

$$a(L)Z_{t,i} = b(L)\varepsilon_{t,i}, \quad \text{for } i = 1, \dots, n; t = 1, \dots, T, \quad (1)$$

where L is a lag operator, such that $LZ_{t,i} = Z_{t,i-1}$, $b(L) = (1 + \Theta_1 L + \dots + \Theta_q L^q)$, $a(L) = (1 - A_1 L - \dots - A_p L^p)$, and $\varepsilon_{t,i}$ denotes the daily structural innovation for the i -th day within month t (with a total of n days). Here, n is the number of daily observations within a month (low-frequency), and we define $y_{t,0} = y_{t-1,n}$ as the transition between months. Our goal is to forecast the temporal aggregate $\bar{Z}_{T+h} = \frac{1}{n} \sum_{i=1}^n Z_{T+h,i}$ (i.e. a low-frequency average) at some future monthly horizon h .

Suppose that all data is available to the practitioner. Then, the bottom-up approach for the period average is the ex-post average of the daily forecasts. This yields the optimal forecast, which we label as Bottom-up MIDAS (BUMIDAS):

Proposition 1. *[BUMIDAS for High-Frequency VARMA(p,q)] The optimal mixed-frequency direct forecast of a temporally aggregated VARMA(p,q) with $n \geq 2$ and forecast horizon h is given by*

$$\mathbb{E}_{T,n} [\bar{Z}_{T+h}] = \sum_{j=1}^p \beta_{h,j} Z_{T,n-j+1} + \sum_{\ell=1}^q \alpha_{h,\ell} \varepsilon_{T,n-\ell+1}, \quad (2)$$

where $\beta_{h,j}$ are functions of the autoregressive matrices $A_1, \dots, A_p$ and $\alpha_{h,\ell}$ are functions of the autoregressive matrices and the moving average coefficients $\Theta_1, \dots, \Theta_q$.

Proof. See Appendix A2.1. □

Proposition 1 reveals two important insights. First, the optimal direct mixed-frequency forecast depends only on the most recent p autoregressive and q moving-average terms of the high-frequency process. Temporal aggregation changes the forecast coefficients but not the dimension of this state; consequently, the number of parameters does not increase with the degree of aggregation. Second, constructing the optimal forecast requires the relevant high-frequency information to be observed or recoverable.

There are two immediate advantages of approaching mixed-frequency modeling using the bottom-up approach. First, the bottom-up approach guarantees efficiency (Kohn, 1982; Lütkepohl, 1986). Second, while weighting functions offer a convenient way to express the information structure, they do not always exhibit finite representations (see, Brewer, 1973; Silvestrini and Veredas, 2008). For example, an AR(2) aggregated to two periods has UMIDAS structure with infinite high-frequency moving average terms. The BUMIDAS form therefore supplies an efficient benchmark against which alternatives can be evaluated.

Our results extend naturally to VAR and VMA dynamics (see Appendix A3.1 and Appendix A3.2), as well as to point-in-time sampling schemes (see Appendix A3.3), including end-of-period observations. In each case, the optimal forecast depends on the same underlying high-frequency state, while only the coefficients change with the sampling rule. Hence, the required state dimension is determined by the high-frequency VARMA(p, q) dynamics rather than by the aggregation or sampling scheme.

The approach also extends to nowcasting when the high-frequency target is observed (Appendix A3.4): after accounting for the observed part of the aggregate, forecasting the remainder requires only the latest p high-frequency observations and q innovations. The coefficients depend on the within-period forecast origin, but incorporating newly available observations does not require estimating additional coefficients.

2.1.1 Illustrative Example

Suppose the underlying daily data follow an AR(1) process,

$$y_{t,i} = \rho y_{t,i-1} + \varepsilon_{t,i}, \quad i = 1, \dots, n. \quad (3)$$

The optimal forecast for any day i within month $T + h$ is $\mathbb{E}_{T,n}[y_{T+h,i}] = \rho^{(h-1)n+i} y_{T,n}$. Following the recursive bottom-up approach, the optimal monthly average forecast is obtained by forecasting all days and then taking their monthly average:

$$\mathbb{E}_{T,n}[\bar{y}_{T+h}] = \frac{1}{n} \sum_{i=1}^n \rho^{(h-1)n+i} y_{T,n} = \beta_h y_{T,n}. \quad (4)$$

The middle expression shows that the optimal recursive forecast projects only on the end-of-month observation $y_{T,n}$. The same forecast can be constructed directly at mixed frequency by projecting the future monthly average on the same high-frequency state $y_{T,n}$ using β_h . Recursive bottom-up forecasting and direct BUMIDAS are equivalent in population: the aggregation frequency and forecast horizon change β_h , but the forecast continues to require only the latest daily information.

2.2 The Value of Missing Data

In practice, a researcher may not observe all the components of a high-frequency process, rendering the specification in Proposition 1 infeasible. Consequently, much of the current MIDAS literature focuses on feasible alternatives that project the low-frequency target onto available data. While these approaches provide practical solutions when the high-frequency process is only partially observed, the extent of the resulting information loss remains less well understood. Comparing these approaches with the complete-information BUMIDAS benchmark provides a natural way to quantify this loss, and from a policy perspective, to assess the value of collecting additional high frequency

data.

For exposition, it suffices to discuss the case of a high-frequency bivariate VAR(1) model of the form:

$$Z_{t,i} = AZ_{t,i-1} + \varepsilon_{t,i}, \quad (5)$$

where $Z_{t,i} = (Y_{t,i}, X_{t,i})'$, $\varepsilon_{t,i} = (\varepsilon_{t,i}^y, \varepsilon_{t,i}^x)'$ and our interest is the h period ahead aggregate, $\bar{Z}_{t+h}$.

Let us denote $\mathcal{F}_{T,n}$ as the information set available to the practitioner at time T, n . Then, the unconditional MSFE associated with the forecast $\hat{\bar{Z}}_{t+h}(\mathcal{F}_{T,n})$ can be decomposed as¹:

Proposition 2. *Let $\mathbb{E}[\bar{Z}_{T+h} \mid \mathcal{F}_{T,n}]$ denote the MSFE-optimal forecast given the information set $\mathcal{F}_{T,n}$, and let $\hat{\bar{Z}}_{T+h}(\mathcal{F}_{T,n})$ denote an arbitrary, potentially non-optimal forecast based on the same information set. Then, the unconditional MSFE associated with the forecast $\hat{\bar{Z}}_{T+h}(\mathcal{F}_{T,n})$ is given by:*

$$\mathbb{E}[MSFE(\bar{Z}_{T+h} \mid \mathcal{F}_{T,n})] = \underbrace{\mathbb{V}[\bar{Z}_{T+h} \mid Z_{T,n}]}_{\text{Irreducible/BUMIDAS Variation}} + \underbrace{\beta_h \mathbb{E}[\mathbb{V}[Z_{T,n} \mid \mathcal{F}_{T,n}]] \beta_h'}_{\text{Information Loss}} + \underbrace{\mathbb{E}[B(\bar{Z}_{T+h} \mid \mathcal{F}_{T,n})]}_{\text{Forecast Inefficiency}}. \quad (6)$$

where $B(\bar{Z}_{T+h} \mid \mathcal{F}_{T,n}) = [\mathbb{E}(\bar{Z}_{T+h} \mid \mathcal{F}_{T,n}) - \hat{\bar{Z}}_{T+h}(\mathcal{F}_{T,n})][\mathbb{E}(\bar{Z}_{T+h} \mid \mathcal{F}_{T,n}) - \hat{\bar{Z}}_{T+h}(\mathcal{F}_{T,n})]'$.

Proof. See Appendix A2.2. □

Indeed, the MSFE can be partitioned into three components. The first is an irreducible variance component, which arises from uncertainty about the future given complete information on the underlying high-frequency state $Z_{t,n}$. The second is an information loss penalty which reflects the extent to which $\mathcal{F}_{T,n}$ fails to recover the predictive content of the underlying high-frequency state

¹Note that this decomposition is not limited to VAR processes. Under the VAR(1) DGP above, $Z_{T,n}$ is a sufficient predictive state and the forecast errors are conditionally homoskedastic. More generally, the decomposition applies whenever the current state contains all information in $\mathcal{F}_{T,n}$ that is relevant for forecasting $\bar{Z}_{T+h}$ and the conditional forecast-error variance is state-independent. This condition is automatically satisfied for a Markov state with homoskedastic innovations.

$Z_{t,n}$. The final term is the squared deviation of the forecast from the MSFE-optimal predictor $\mathbb{E}[\bar{Z}_{T+h}|\mathcal{F}_{T,n}]$, and therefore captures losses arising from inefficient use of the information. An immediate consequence of this proposition is that, under the BUMIDAS structure and complete information, the MSFE is minimized:

Corollary 1. For the high-frequency VAR(1), the BUMIDAS specification under complete-information

$\mathcal{F}_{T,n}^C = \sigma(Z_{t,n})$ yields the lower bound on the MSFE such that:

$$\mathbb{E}[MSFE(\bar{Z}_{t+h}|\mathcal{F}_{T,n}^C)] = \mathbb{V}[\bar{Z}_{t+h}|Z_{t,n}] \leq \mathbb{E}[MSFE(\bar{Z}_{t+h}|\mathcal{F}_{T,n})], \quad (7)$$

for any alternative information set $\mathcal{F}_{T,n}$.

This explains why the first term in the decomposition is labelled as “BUMIDAS” variation. More generally, Proposition 2 shows that forecast accuracy can be improved along two distinct margins. First, the information set can be enriched so that it closely captures the informational content of the underlying state. Second, conditional on a given information set, the forecasting rule can be improved so that it more closely approximates the MSFE-optimal predictor.

To further illustrate these ideas, consider forecasts using only aggregate data. A standard approach would be to perform a linear projection on the low-frequency information set $\mathcal{F}_T^{LF} = \sigma(\bar{Y}_T, \dots, \bar{X}_T, \dots)$. Such forecasts however, discard the useful and available high frequency information in $X_{t,n}$, motivating the large literature on mixed frequency forecasting. Indeed, mixed frequency methods instead retain high frequency information whenever possible, and therefore condition on a richer information set $\mathcal{F}_T^{MF} = \sigma(\bar{Y}_T, \dots, X_{t,n}, \dots)$. The gains due to moving from $\mathcal{F}_T^{LF}$ to $\mathcal{F}_{T,n}^{MF}$ reflect the reduction to the information-loss margin of Proposition 2.

Conditional on a particular information set however, there remains a separate question of how efficiently the available information set is used. For example, methods such as UMIDAS, RMIDAS

and state space approaches will impose different structures (i.e. functional specification, lag selection lengths) on $\mathcal{F}_T^{MF}$. Therefore, they differ in how closely they approximate the MSFE-optimal predictor. Improvements obtained by changing the forecasting rule while holding the information set fixed operate along the inefficiency margin.

2.3 BUMIDAS with a Proxy Variable

While mixed-frequency methods can improve upon low-frequency approaches by making better use of the observed high-frequency data, the associated information set is typically motivated by data availability rather than chosen for its optimality. In many applications, observable proxies may provide a more informative representation of the missing high-frequency state, thereby improving upon the existing mixed-frequency methods. For exposition, consider the mixed-frequency information set $\mathcal{F}_T = \sigma(X_{T,n}, \dots, \bar{Y}_T, \dots, Y_{T,n}^*, \dots)$, where $Y_{t,i}^*$ is a proxy variable relating to $Y_{t,i}$ such that:

$$Y_{t,i}^* = BY_{t,i} + \nu_{t,i}. \quad (8)$$

Here, B is the strength of the proxy and $\nu_{t,i}$ is an i.i.d. noise (i.e. the measurement error) with 0 mean and variance σ_ν^2 . To understand the structure of the optimal forecast for $\bar{Y}_{t+h}$ using $\mathcal{F}_T$, we formalize our setting into a state space framework (for complete technical details and derivations, please see Appendix A2.3). Let $\tilde{Z}_T$ be the $2n \times 1$ stacked vector of the high-frequency observations $Z_{T,i}$ within period T , with associated stacked innovations $\tilde{u}_T$, and let W_T be the $(2n+1) \times 1$ vector containing the aggregate $\bar{Y}_T$, the N stacked high-frequency observations $\tilde{X}_T$, and the N stacked high-frequency proxy observations $\tilde{Y}_T^*$. The stacked noise term $\tilde{\nu}_T$ is also a $(2n+1) \times 1$ vector with 0's except for the entries corresponding to the n proxies. Finally, $\mathcal{A}$ and C are the autoregressive and measurement coefficient matrices respectively, stacked in such a way that the dimensions of

the following model agree:

$$\begin{aligned}\tilde{Z}_T &= \mathcal{A}\tilde{Z}_{T-1} + \tilde{u}_T, \quad (\text{State Equation}) \\ W_T &= C\tilde{Z}_T + \tilde{v}_T. \quad (\text{Measurement Equation})\end{aligned}\tag{9}$$

This formulation coincides with the standard state space representation as discussed in Gourieroux and Monfort (1997), Chapter 15.1. Then, standard filtering equations can be applied. Under this model, the filtered forecast $\hat{\tilde{Z}}_{T+h|T} = \mathbb{E}[\tilde{Z}_{T+h}|\mathcal{F}_T]$ admits an infinite representation of the form (see Appendix A2.4):

$$\hat{\tilde{Z}}_{T+h|T} = \sum_{j=0}^{\infty} \Theta_{T,j} W_{T-j},\tag{10}$$

where $\Theta_{T,j} = (\Theta_{T,j}^{\tilde{Y}}, \Theta_{T,j}^X, \Theta_{T,j}^{Y^*})$ are the weights on observations $W_{T-j} = (\bar{Y}_{T-j}, \tilde{X}_{T-j}, \tilde{Y}_{T-j}^*)'$. Bai et al. (2013) derive a related infinite representation from a periodic high-frequency state-space model and, under steady state, show how the resulting Kalman weights can be linked to parametric MIDAS lag structures. They also show that reduced-form MIDAS regressions can perform comparably to the corresponding state-space forecasts. Their results highlight a trade-off: MIDAS can sacrifice some efficiency under correct state-space specification while being more robust to misspecification or overparameterization.

We retain this reduced form insight, but our setup here does not impose a particular parametric structure on the implied weights. Instead, our interest is to use our representation to study how those weights change with proxy quality.

Proposition 3. *Under the state space model in (9) and its associated infinite recursive representation in (A2.4):*

1. *If the proxy strength $B \rightarrow 0$ or proxy noise $\sigma_v^2 \rightarrow \infty$ (the proxy is uninformative), then:*

- *(Zero weighting) $\Theta_{T,j}^{Y^*} \rightarrow \begin{smallmatrix} 0 \\ (2n \times n) \end{smallmatrix}$.*

2. If the proxy strength $B \neq 0$ and the proxy noise $\sigma_\nu^2 \rightarrow 0$ (the proxy is perfect), then:

- (Redundancy of Aggregate) $\mathcal{F}_T = \mathcal{F}_T^r$, where $\mathcal{F}_T^r = \sigma(X_{T,n}, \dots, Y_{T,n}^*, \dots)$.
- (No information loss) $\Sigma_{T|T} = \mathbb{V}[\tilde{Z}_T | \mathcal{F}_T] = \begin{smallmatrix} 0 \\ 2n \times 2n \end{smallmatrix}$.
- (BUMIDAS Specification) $\hat{\tilde{Z}}_{T|T} = \Theta_{T,0}^X \tilde{X}_T + \Theta_{T,0}^{Y^*} \tilde{Y}_T^*$.

Proof. See Appendix A2.5 □

This proposition characterizes the two limiting cases of proxy quality. When the proxy is uninformative, its associated weights vanish and the projection depends only on the low-frequency aggregate $\bar{Y}$ and the collection of observed high-frequency covariates X . Thus, it reverts to the UMIDAS/RMIDAS information structure. In the other extreme, a perfect proxy renders the aggregate redundant because it fully reveals the underlying high-frequency target. In this case, the filtering covariance $\Sigma_{T|T}$, which is the analogue of the information loss term in Proposition 2, vanishes. Thus, the complete information benchmark is attained. Moreover, all lagged reduced form weights disappear, leaving a projection based on only the current high-frequency state. This corresponds to the BUMIDAS structure, where the most recent state is sufficient under the Markov dynamics. In the simulations below, we show how intermediate levels of proxy quality move the forecasting specification between these two limiting cases.

3 Simulation Evidence

We use Monte Carlo simulations to evaluate specification selection and out-of-sample forecast accuracy when the high-frequency target is observed, observed through a proxy, or unavailable. We consider three daily calibrations: a persistent triangular VAR(1) in which the first variable depends solely on its lag and affects the second contemporaneously and with lag, but not vice versa; a feedback VAR(1) with two-way dependence; and a VAR(3) with richer own- and cross-variable

dynamics. The daily processes are aggregated to weekly, monthly, and quarterly frequencies. Appendix A4.1 reports complete details.

3.1 Specification Evidence

We first examine whether standard specification tests recover the parsimonious mixed-frequency structure implied by BUMIDAS when the high-frequency state is observed. Our interest is whether the latest p high-frequency observations are relevant and whether additional structures commonly used in temporally aggregated and mixed-frequency models are needed: earlier high-frequency observations (Foroni et al., 2015a), aggregate autoregressive terms (Weiss, 1984), and residual serial correlation that would motivate a low-frequency moving-average component (Foroni et al., 2019).

We test the immediately preceding high-frequency state rather than the full unrestricted lag block; details in Appendix A4.2. This directly tests the relevant state restriction while avoiding the finite-sample size distortions and loss of power that can arise when many high-frequency coefficients are tested jointly (Ghysels et al., 2020). Conventional F-tests using larger high-frequency lag blocks give similar results in larger samples.

Table 1. Specification Tests with the Observed High-Frequency State

Target	20 Years				40 Years			
	BUMIDAS state	Extra HF state	LF AR state	Residual serial corr.	BUMIDAS state	Extra HF state	LF AR state	Residual serial corr.
Monthly								
Univariate, y_1	1.000	0.053	0.051	0.049	1.000	0.054	0.048	0.050
Multivariate, y_2	1.000	0.055	0.049	0.050	1.000	0.052	0.049	0.052
Quarterly								
Univariate, y_1	1.000	0.052	0.045	0.042	1.000	0.046	0.047	0.050
Multivariate, y_2	1.000	0.050	0.050	0.045	1.000	0.049	0.050	0.049

Notes: Rejection frequencies from 5,000 simulations at the 5 percent level for the triangular VAR(1). “Extra HF state” tests the preceding high-frequency state; “LF AR state” tests lagged aggregate information; “Residual serial corr.” is a first-order Breusch–Godfrey test.

Table 1 supports the BUMIDAS structure when the high-frequency state is observed. The last

p high-frequency lags are relevant in every simulation, while rejection frequencies for additional high-frequency information, aggregate AR terms, and residual serial correlation remain close to the nominal 5 percent level. Thus, only the terminal high-frequency state is needed, and its sufficiency does not depend on the degree of temporal aggregation. Weekly aggregation, alternative DGPs, and end-of-period sampling, reported in Appendix A4.2, yield the same conclusions.

Table 2. BIC Specification Selection with the Observed High-Frequency State

Target	20 Years		40 Years	
	Modal specification	Selection frequency	Modal specification	Selection frequency
Monthly				
Univariate, y_1	(1, 0)	0.956	(1, 0)	0.971
Multivariate, y_2	(1, 1)	0.951	(1, 1)	0.949
Quarterly				
Univariate, y_1	(1, 0)	0.892	(1, 0)	0.945
Multivariate, y_2	(1, 1)	0.891	(1, 1)	0.938

Notes: Entries report the modal BIC specification and its selection frequency across 5,000 simulations. (q_1, q_2) denotes the number of high-frequency observations of y_1 and y_2 . The BUMIDAS specifications are (1, 0) for y_1 and (1, 1) for y_2 .

When the underlying DGP is unknown, the relevant high-frequency state must be selected empirically. We therefore use AIC, BIC, and HQIC to jointly search over the number of high-frequency lags for each variable, where (q_1, q_2) denotes the numbers of y_1 and y_2 observations retained. The search also allowed for inclusion of aggregate AR and MA(1) terms, but they were found to be unnecessary, so they are omitted for brevity. Table 2 reports BIC for the triangular design; AIC, HQIC, mean lag counts, weekly aggregation, and alternative parameterization results demonstrate consistent results and are reported in Appendix Table A2.

BIC reliably recovers the parsimonious BUMIDAS state, selecting the correct specification in over 90 percent of simulations for monthly and quarterly data with a 40 year sample. Consistent with the VAR lag-selection literature (Lütkepohl, 1985; Hacker and Hatemi-J, 2008), expanded results (Appendix A4.2) demonstrate that AIC retains unnecessary high-frequency observations,

with HQIC generally in between. In the quarterly feedback design, however, weak cross-variable effects are frequently omitted, particularly in shorter samples, and HQIC performs relatively better than BIC. The practical guidance is therefore to use BIC for weekly and monthly data, with HQIC preferable for quarterly data in smaller samples with weak cross-variable effects. Consistent with Gonzalo and Pitarakis (2002), AIC becomes relatively more attractive as the parameter space expands in higher-dimensional systems.

Consider now the case when the high-frequency path of the target is unobserved but that a noisy high-frequency proxy, $y_{t,i}^*$, is available as in equation 8. Strong, moderate, and weak proxies are reported with $(B, \sigma_\nu/\sigma_Y)$ equal to $(0.75, 0.25)$, $(0.50, 0.50)$, and $(0.25, 0.75)$, yielding proxy–target correlations of 0.949, 0.707, and 0.316, respectively. For a univariate target, the available high-frequency information consists of the target proxy. For the multivariate target, the high-frequency predictor remains observed while the high-frequency target is proxied. Alternative criteria, and parameterizations, are provided in Appendix A4.3.

In the presence of a proxy, the specification tests in Table 1 generally reject the joint exclusion of the latest p high-frequency observations. However, exclusion of earlier high-frequency observations is also frequently rejected, even with a strong proxy, especially at monthly and weekly frequencies. As proxy quality deteriorates, rejection of the exclusion of aggregate AR terms becomes more pervasive. Rejection of the exclusion of earlier high-frequency observations can nevertheless fall when proxy quality is weak in finite samples, because lagged proxy observations also become less informative as measurement error increases. Rejection frequencies are generally highest at the weekly frequency and lowest under quarterly aggregation.

Motivated by these results, we use BIC to select the finite unrestricted specification that best approximates the latent-state filter when the target is observed at high frequency only through a proxy. Table 3 reports results for the second equation of the multivariate VAR(1) with two-way

dependence; results for additional DGPs are reported in Appendix A4.3. In each replication, a global grid search jointly selects the number of recent high-frequency observations of each variable and the aggregate AR order, allowing either high-frequency variable to be excluded.

Table 3. BIC Specification Selection with High-Frequency Proxies

Aggregation	Proxy quality	Corr. (y^*, y)	Modal specification [frequency]	Mean selected
Quarterly	Strong	0.949	(1, 1, 0, 0) [41]	(1.06, 0.98, 0.16, 0.00)
	Moderate	0.707	(1, 0, 0, 0) [59]	(1.10, 0.18, 0.27, 0.01)
	Weak	0.316	(1, 0, 0, 0) [64]	(1.11, 0.05, 0.29, 0.01)
Monthly	Strong	0.949	(1, 2, 1, 0) [30]	(1.06, 2.40, 0.92, 0.07)
	Moderate	0.707	(1, 0, 2, 0) [28]	(1.05, 0.54, 1.47, 0.19)
	Weak	0.316	(1, 0, 2, 0) [38]	(1.05, 0.05, 1.49, 0.21)
Weekly	Strong	0.949	(1, 1, 2, 0) [56]	(1.07, 1.80, 2.11, 0.03)
	Moderate	0.707	(1, 0, 2, 0) [50]	(1.08, 0.37, 2.15, 0.01)
	Weak	0.316	(1, 0, 2, 0) [76]	(1.08, 0.03, 2.15, 0.00)

Notes: Modal specifications are followed by their percentage selection frequencies in brackets. “Mean selected” reports the mean selected specification across 5,000 simulations using 40 years of data and the second variable in the feedback VAR. A specification $(q_1, q_2, p_{AR}, q_{MA})$ denotes the numbers of high-frequency predictor and target-proxy observations and the aggregate AR and MA terms.

There are three key takeaways.

First, even strong proxies can lead BIC to select information beyond the most recent p high-frequency observations. At the quarterly frequency, the modal specification retains the current observed predictor and target proxy. At the monthly frequency, it also includes an earlier proxy observation and an aggregate AR term.

Second, weaker proxies shift the specification toward aggregate target history, rather than simply toward more proxy lags. At monthly and weekly frequencies, the modal specifications exclude moderate and weak proxies while retaining the current observed predictor and two aggregate AR terms. Thus, aggregate target history can substitute for noisy high-frequency target information without materially extending the observed predictor’s high-frequency history.

Third, richer specifications are selected at higher observation frequencies. Quarterly specifi-

cations remain the most parsimonious, with the modal specification retaining only the current observed predictor when the proxy is moderate or weak. Monthly and weekly modal specifications instead include one or two aggregate AR terms. Thus, greater temporal aggregation does not mechanically imply a richer specification, and the selected low-frequency dynamics need not mirror the VAR(1) order of the underlying high-frequency process.

Overall, the results support the latent-state interpretation of BUMIDAS. When the target is proxied, BIC selects among high-frequency observations and aggregate terms to approximate the latent state, rather than mechanically moving toward a longer UMIDAS lag structure as proxy quality deteriorates.

3.2 Out-of-Sample Forecast Performance

We compare out-of-sample forecast performance across mixed-frequency methods and report mean-squared forecast error (MSFE) ratios relative to the period-average no-change forecast. Mean directional accuracy is reported in Appendix A4.5, with details of the forecast evaluation in Appendix A5.1. Bracketed entries give the percentage of Monte Carlo replications in which a one-sided Diebold–Mariano test (Diebold and Mariano, 1995) rejects equal predictive accuracy at the 10% level in favor of BUMIDAS-BIC. The tests use squared-error loss differentials and Newey–West HAC standard errors (Newey and West, 1987).

The simulated daily DGPs are bivariate triangular VAR(1) or VAR(3). The forecast target is the one-period-ahead average of the second variable. Table 4 reports forecast accuracy when the high-frequency target is observed, while Table 5 considers the case in which a noisy high-frequency proxy for the target is available. The end-of-period (EoP) no-change forecast is included as a naive benchmark since it is the optimal forecast of the period average under a random walk null (Ellwanger and Snudden, 2023; McCarthy and Snudden, 2025). Bottom-up and BUMIDAS condition on the

last p daily observations, as in Proposition 1, whereas BUMIDAS-BIC selects the high-frequency lag order at each forecast origin using BIC. UMIDAS and RMIDAS project on low-frequency lags of X and high-frequency Y .

Table 4. Forecast Performance when High-Frequency Data of the Target is Observed

Method	Weekly	Monthly	Quarterly
<i>Panel A. VAR(1)</i>			
BU-Recursive	0.658	0.560	0.535
BUMIDAS	0.658	0.562	0.546
BUMIDAS-BIC	0.658	0.562	0.548
UMIDAS	0.835 [100]	0.679 [88]	1.258 [97]
RMIDAS	0.839 [100]	0.693 [91]	0.657 [57]
LF VAR	0.967 [100]	0.982 [100]	0.967 [94]
EoP NC	0.710 [96]	0.621 [68]	0.599 [42]
<i>Panel B. VAR(3)</i>			
BU-Recursive	0.700	0.577	0.536
BUMIDAS	0.702	0.585	0.564
BUMIDAS-BIC	0.704	0.585	0.549
UMIDAS	0.812 [100]	0.788 [96]	9.370 [100]
RMIDAS	0.807 [100]	0.664 [73]	0.601 [27]
LF VAR	0.811 [100]	0.717 [88]	0.633 [38]
EoP NC	1.131 [100]	1.395 [100]	2.245 [100]

Notes: MSFE ratios relative to the period-average no-change forecast (mean of 5,000 replications, $T = 40$ years). Bracketed values are percentage rejection frequencies of a one-sided DM test at the 10% level in favor of BUMIDAS-BIC. The daily target follows a VAR(1) or VAR(3), aggregated to $n \in \{5, 21, 63\}$. BU-Recursive averages daily-VAR forecasts; LF VAR and EoP NC denote the low-frequency VAR and end-of-period no-change forecasts, respectively.

Table 4 establishes the central result. BU-Recursive, BUMIDAS, and BUMIDAS-BIC deliver essentially the same forecast accuracy and have lower MSFE ratios than the alternatives across both DGPs and all aggregation frequencies. The near equivalence of the direct BUMIDAS and recursive bottom-up forecasts is consistent with Proposition 1. BUMIDAS-BIC also retains nearly all of these gains when the lag order is selected rather than known. The gains are substantial relative to the period-average no-change forecast and increase from weekly to monthly and quarterly aggregations.

The forecasts clarify the source of these differences. UMIDAS becomes unstable under quarterly aggregation, especially for the VAR(3). RMIDAS avoids this severe deterioration but remains less

accurate than BUMIDAS, consistent with its omission of high-frequency target information.

The DGP also determines the value of the benchmarks. Under the VAR(1), persistence makes the end-of-period observation informative. Under the VAR(3), however, the end-of-period observation no longer summarizes the relevant daily lags, and its naive forecast performance deteriorates sharply. The low-frequency VAR becomes relatively more competitive under the VAR(3) because aggregate lags better capture the richer dynamics, but it continues to lose information through temporal aggregation. Appendix Table A9 shows that directional accuracy produces the same broad ranking.

Equal predictive accuracy is rejected in favor of BUMIDAS-BIC in 88–100% of replications against UMIDAS, with frequent rejections also against the LF VAR and EoP NC. Against RMIDAS, rejection frequencies are 100% weekly, 73–91% monthly, and 27–57% quarterly. The larger number of out-of-sample observations in weekly designs may contribute to their higher rejection frequencies.

Next, we consider environments in which the high-frequency target X is unobserved or observed only through a proxy, as characterized in Section 2.3. The available information comprises high-frequency Y , low-frequency lags of X , and, when available, the high-frequency proxy X^* . Without a proxy, this is the conventional information environment for UMIDAS and RMIDAS.

BUMIDAS-BIC again selects among the available high- and low-frequency terms to extract a finite reduced-form approximation of the filter. To assess the cost of restricting the projection to the most recent daily observations, we also report a Last- p specification that projects on the last p observations of each available high-frequency variable under the known DGP order. Thus, Last- p includes Y and X^* when a proxy is available and Y and $\bar{X}$ when it is unavailable.

To favor UMIDAS and RMIDAS, we assume that the DGP order is known. In both tables, UMIDAS includes $p(n-1)$ unrestricted high-frequency lags of Y , whereas RMIDAS applies Almon restrictions to $n-1$ high-frequency lags of Y . When X^* is available, both specifications are

extended to include the corresponding proxy history. Both methods also include p low-frequency autoregressive lags of X .

Table 5. Forecast Performance when the High-Frequency Target is Proxied or Unavailable

Frequency	Proxy (Corr.)	Last- p	BUMIDAS BIC	UMIDAS	RMIDAS	LF VAR
<i>Panel A. VAR(1)</i>						
Quarterly	Strong (0.95)	0.564	0.568	31.3 [100]	0.660 [45]	0.966 [91]
	Moderate (0.71)	0.572	0.576	30.5 [100]	0.670 [46]	0.966 [90]
	Weak (0.32)	0.572	0.576	30.6 [100]	0.672 [47]	0.966 [90]
	Unavailable	0.573	0.574	1.258 [97]	0.657 [47]	0.966 [92]
Monthly	Strong (0.95)	0.635	0.629	0.687 [53]	0.662 [34]	0.982 [100]
	Moderate (0.71)	0.662	0.656	0.718 [52]	0.693 [37]	0.982 [99]
	Weak (0.32)	0.664	0.658	0.722 [55]	0.697 [40]	0.982 [99]
	Unavailable	0.652	0.652	0.679 [34]	0.693 [44]	0.982 [100]
Weekly	Strong (0.95)	1.065	0.812	0.808 [7]	0.814 [18]	0.967 [100]
	Moderate (0.71)	1.213	0.827	0.835 [31]	0.838 [39]	0.967 [100]
	Weak (0.32)	1.225	0.829	0.837 [35]	0.840 [41]	0.967 [100]
	Unavailable	0.872	0.839	0.835 [0]	0.839 [0]	0.967 [100]
<i>Panel B. VAR(3)</i>						
Quarterly	Strong (0.95)	0.567	0.554	6.2 [100]	0.607 [36]	0.633 [46]
	Moderate (0.71)	0.575	0.565	6.0 [100]	0.609 [32]	0.633 [42]
	Weak (0.32)	0.585	0.572	6.3 [100]	0.615 [31]	0.633 [38]
	Unavailable	0.576	0.569	9.4 [100]	0.601 [25]	0.633 [38]
Monthly	Strong (0.95)	0.589	0.591	0.895 [99]	0.614 [32]	0.716 [86]
	Moderate (0.71)	0.624	0.630	0.938 [99]	0.632 [13]	0.716 [71]
	Weak (0.32)	0.664	0.656	0.982 [99]	0.660 [15]	0.716 [61]
	Unavailable	0.656	0.654	0.788 [85]	0.664 [21]	0.716 [64]
Weekly	Strong (0.95)	0.724	0.723	0.726 [16]	0.721 [6]	0.811 [99]
	Moderate (0.71)	0.883	0.787	0.785 [8]	0.780 [2]	0.811 [62]
	Weak (0.32)	1.069	0.806	0.814 [29]	0.804 [7]	0.811 [22]
	Unavailable	0.813	0.813	0.812 [8]	0.807 [2]	0.811 [7]

Notes: Mean MSFE ratios relative to the period-average no-change forecast (5,000 replications, $T = 40$ years). Bracketed values are percentage rejection frequencies of a one-sided DM test at the 10% level in favor of BUMIDAS-BIC. The daily target follows a VAR(1) or VAR(3), aggregated to $n \in \{5, 21, 63\}$. Corr. denotes the correlation between the high-frequency proxy and the latent daily target. Bold indicates the lowest MSFE among BUMIDAS-BIC, UMIDAS, RMIDAS, and the LF VAR; Last- p is excluded as a known-order benchmark. Unavailable denotes that neither daily X nor its proxy X^* is observed.

When the high-frequency target is unavailable, BUMIDAS-BIC has lower MSFE than UMIDAS, RMIDAS, and the LF VAR at monthly and quarterly frequencies under both DGPs. The gains are largest relative to UMIDAS under quarterly aggregation, where the unrestricted specification be-

comes severely overparameterized, particularly under the VAR(3). The DM test frequently rejects equal predictive accuracy in favor of BUMIDAS-BIC, especially against UMIDAS at the monthly and quarterly frequencies. RMIDAS avoids the extreme quarterly deterioration of UMIDAS and is the closest conventional MIDAS competitor, especially under the VAR(3). At the weekly frequency, however, UMIDAS and RMIDAS are marginally more accurate, and the DM tests provide essentially no evidence in favor of BUMIDAS-BIC. Thus, the value of selecting a parsimonious approximation is greatest when the frequency mismatch or the underlying lag order is large.

The proxy results show that additional high-frequency information and forecast performance need not move monotonically together in finite samples. A strong proxy generally improves BUMIDAS-BIC and recovers much of the observed-target forecast gain at the monthly and quarterly frequencies, whereas weak proxies frequently produce MSFE ratios similar to those obtained when no proxy is available. The cost of restricting the projection to the latest p high-frequency observations is clearest at the weekly frequency: Last- p deteriorates as the proxy weakens, whereas BUMIDAS-BIC limits these losses through a search that permits proxy exclusion and additional high- and low-frequency terms. For UMIDAS, adding the proxy can substantially worsen forecast accuracy because it introduces many additional unrestricted coefficients, whereas RMIDAS and BUMIDAS-BIC limit this estimation cost. Thus, the information gained from a proxy can be offset by the finite sample cost of estimating the expanded specification.

In summary, when the daily target is observed, BUMIDAS implements the first-best direct specification and closely matches BU-Recursive. When the target is proxied or unavailable, the relevant high-frequency state must be inferred, motivating the unrestricted BIC-selected approximation.

3.3 Quantifying Information Environments

To approximate the information-loss term in Proposition 2, we use prespecified projections estimated over 100,000 low-frequency periods, reducing finite-sample estimation effects and avoiding model selection. We consider four information environments: low-frequency data only; the aggregate target and high-frequency predictors; this mixed-frequency setting augmented with a high-frequency target proxy; and complete high-frequency information.

We forecast $\bar{Y}_{T+h}$, the aggregate of the first variable in the bivariate VAR(3) of the previous section and Appendix A4.1. Using the daily vector $Z_{t,i}$, let $Z_{T,n}^{(3)} = (Z'_{T,n}, Z'_{T,n-1}, Z'_{T,n-2})'$ denote the complete predictive state and $\mathcal{F}_{T,n}^C = \sigma(Z_{T,n}^{(3)})$ its information set. Let β_h contain the coefficients on $Z_{T,n}^{(3)}$ in the complete-information forecast of $\bar{Y}_{T+h}$. Following Proposition 2, information loss is measured by:

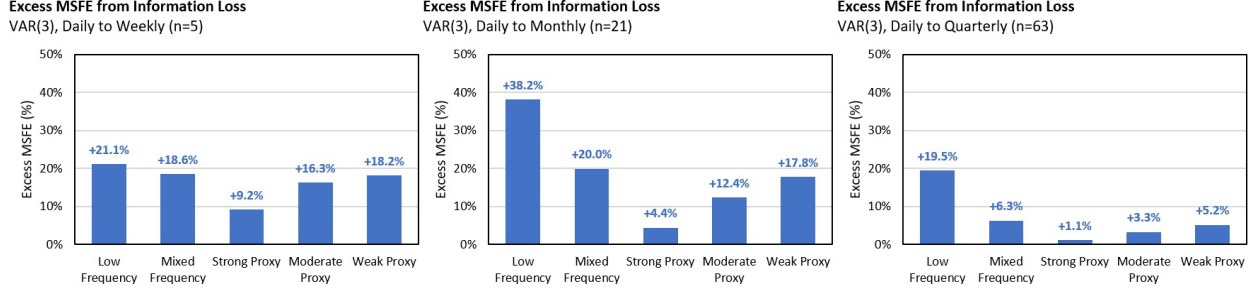
$$\text{MSFE}_{\text{Excess}}(\mathcal{F}_{T,n}) = \frac{\beta_h \mathbb{E} \left[\mathbb{V} \left(Z_{T,n}^{(3)} \mid \mathcal{F}_{T,n} \right) \right] \beta_h'}{\mathbb{V} \left(\bar{Y}_{T+h} \mid \mathcal{F}_{T,n}^C \right)} \times 100\%. \quad (11)$$

The ratio expresses the information-loss term as a percentage of the irreducible complete-information forecast variance and is zero when the complete predictive state is observed. In the simulations, we approximate this population ratio by projecting $\bar{Y}_{T+h}$ directly onto each prespecified finite lag set, subtracting the MSFE of its projection onto $Z_{T,n}^{(3)}$, and dividing by that complete-information MSFE.

Under incomplete information, the reduced-form filter generally depends on an infinite history, as shown in equation (A2.4). We approximate each information environment using a rich but finite unrestricted projection. The low-frequency specification includes $3n$ lags of both $\bar{Y}$ and $\bar{X}$. The standard mixed-frequency specification includes $3n$ lags of $\bar{Y}$ together with $p(n-1)$ high-frequency observations of X , with $p = 3$. The proxy specification additionally includes $p(n-1)$ high-frequency observations of Y^* . The complete-information specification contains the p most recent

high-frequency observations of both Y and X , as in BUMIDAS. We calculate the excess-MSFE ratio separately in each of 1,000 Monte Carlo replications and report the average across replications. Expanding these lag windows further has a negligible effect on the estimated information-loss ratios.

Figure 1. Information Loss Across Information Environments



Note: Bars show estimated information loss as a percentage of the complete-information BUMIDAS MSFE. Lower bars indicate less information loss; zero corresponds to complete information. The information sets contain low-frequency aggregates only, the aggregate target and high-frequency predictors, or the latter augmented with a target proxy. Results are for a daily VAR(3) aggregated to $n \in \{5, 21, 63\}$, averaged over 1,000 Monte Carlo replications with 100,000 low-frequency observations.

The information-loss ratios in Figure 1 show that, under daily-to-weekly aggregation, the standard mixed-frequency information set improves only slightly on the low-frequency set, reducing excess MSFE from 21.1% to 18.6%. The gains are larger under monthly aggregation, from 38.2% to 20.0%, and quarterly aggregation, from 19.5% to 6.3%. Thus, observing high-frequency predictors eliminates a growing share of the loss associated with low-frequency information. However, the level of information loss peaks under monthly aggregation for both information sets.

The value of the proxy also varies with the degree of aggregation. At the weekly frequency, a strong proxy is needed to achieve more than a modest improvement over the standard mixed-frequency information set. At the monthly and quarterly frequencies, even moderate and weak proxies reduce the remaining loss more noticeably. A strong proxy reduces excess MSFE from 18.6% to 9.2% weekly, from 20.0% to 4.4% monthly, and from 6.3% to 1.1% quarterly, bringing the forecast closest to the complete-information benchmark in each case.

Overall, the results show that information loss can remain substantial when relevant high-

frequency information is unavailable. Changing lag structures or forecasting specifications alone cannot eliminate the loss due to unrecoverable state information from the available history.

4 Applications

The empirical applications are organized around whether the high-frequency path of the target is observed, proxied, or unavailable. The first forecasts interest rates, a univariate setting with an observed target. The second forecasts exchange rates using oil as a predictor, a multivariate setting with an observed target. The third forecasts inflation, where the target is proxied, and the fourth nowcasts GDP, where it is unavailable.

4.1 3-Month Treasury Bill

We analyze real-time monthly and quarterly average level forecasts of the nominal short-term interest rate on the U.S. 3-Month Treasury Bill. Daily closing yields are obtained from the FRED database of the Federal Reserve Bank of St. Louis (ticker: DGS3MO). Period averages are calculated as the simple average of daily closing yields. The time series begins in January 1990, with the forecast evaluation sample running from January 2000 to June 2025.

We compare recursive bottom-up forecasts with four direct specifications: a model based on period averages, BUMIDAS, an unrestricted specification using $p(n-1)$ daily lags (Extra HF), and an Almon-restricted specification using $n-1$ daily lags (Almon).² The latter two apply conventional MIDAS lag structures to the target’s own daily history. We use $p = 1$, consistent with the BIC from the daily data estimated on the first difference of the rate, with $n = 21$ and $n = 63$, corresponding to the average number of trading days in the period. Forecasts for the nominal level on day i of period $t + h$, conditional on information available at period t , are based on the model-implied

²Regardless of starting values or estimation methods, Beta restrictions failed to converge during at least one period of the expanding window estimation and are therefore only Almon restrictions are reported.

cumulative sum between day t, n and day $t + h, i$. These daily forecasts are then averaged to obtain the period forecast. BUMIDAS uses the same number of lags as the daily bottom-up model.

For comparability with previous studies, we report MSFE ratios relative to the period-average no-change forecast, with one-sided Diebold–Mariano p -values in parentheses. Directional success ratios and tests against the end-of-period no-change forecast are reported in Appendix A5.2.

Table 6. Model-based Forecasts of the Nominal 3-Month Treasury Bill Rate

Method	Direct	Bottom-up	BUMIDAS	Extra HF	Almon
Data	Average	Daily	Mixed	Mixed	Mixed
horizon	Monthly Forecasts				
1	1.00 (0.535)	0.56 (0.000)	0.56 (0.000)	0.68 (0.006)	0.66 (0.002)
3	0.99 (0.382)	0.82 (0.000)	0.81 (0.002)	0.87 (0.058)	0.79 (0.009)
12	0.91 (0.083)	0.95 (0.000)	0.87 (0.019)	0.88 (0.098)	0.84 (0.005)
horizon	Quarterly Forecasts				
1	0.99 (0.435)	0.40 (0.001)	0.41 (0.002)	1.83 (0.898)	0.50 (0.013)
2	0.96 (0.311)	0.69 (0.002)	0.67 (0.004)	1.95 (0.979)	0.67 (0.013)
4	0.91 (0.171)	0.84 (0.001)	0.78 (0.015)	1.85 (0.987)	0.70 (0.015)

Note: Real-time, out-of-sample forecasts of the nominal monthly and quarterly average 3-month Treasury Bill rate, ending in 2025M6. MSFE ratios are relative to the corresponding period-average no-change forecast, with one-sided Diebold–Mariano p -values reported in parentheses.

Table 6 shows that the bottom-up recursive and BUMIDAS forecasts perform similarly, particularly at short horizons. Relative to the direct forecast based on period averages, both reduce MSFE by approximately 44 percent at one month and 60 percent at one quarter. At one month, BUMIDAS reduces the MSFE ratio by 0.12 relative to Extra HF and 0.10 relative to Almon, with both differences significant at the 10 percent level ($p = 0.085$ and $p = 0.068$, respectively). The cost of overparameterization becomes pronounced under quarterly aggregation: Extra HF produces MSFE ratios above 1.8 and performs worse than the no-change benchmark at every horizon. At longer horizons, BUMIDAS and Almon generally produce the lowest MSFEs and improve on both Extra HF and the bottom-up recursive forecast.

The forecast rankings remain consistent across the sample, including during COVID-19 (Ap-

pendix Figure A1). They are also robust to drift terms, seasonal adjustment, and alternative information criteria, where increasing p worsens forecast accuracy, especially with $p(n - 1)$ unrestricted terms. Overall, the application confirms that the first-order driver of the results is the efficient use of the high-frequency data of the target.

4.2 Exchange rate forecasts

As a second application, we conduct out-of-sample forecasts over 2000M1–2025M6 of monthly-average CAD/USD and NOK/USD exchange rates using daily WTI oil prices, with data beginning in 1986M1 (see Data Appendix A1). The exercise is motivated by evidence that oil–exchange rate predictability is concentrated at high frequencies and weakened by temporal aggregation (Ferraro et al., 2015; Foroni et al., 2015b). While our baseline forecasts monthly averages, Appendix A5.3 repeats the exercise for end-of-period exchange rates common in the literature. The daily model uses log changes in nominal exchange rates and oil prices, with BIC selecting a daily VAR(1) for both currencies. Consistent with the daily VAR, BUMIDAS projects on the most recent daily exchange rate and oil observations. UMIDAS and RMIDAS instead project on monthly exchange rates using one month of unrestricted or Almon-weighted daily oil lags. Results are reported relative to the monthly-average no-change forecast, with a VAR estimated using monthly-average data and the end-of-month no-change forecast included as benchmarks.

Table 7 shows substantial gains from retaining the high-frequency information on the target. BUMIDAS and the recursive bottom-up forecast perform similarly and reduce MSFE by 30–40 percent relative to the monthly-average no-change benchmark. In contrast, the low-frequency VAR, UMIDAS, and RMIDAS discard the most recent exchange-rate observation by conditioning on the monthly average. Their MSFE ratios remain around 0.9–1.0, and BUMIDAS significantly outperforms all three. The bottom-up forecasts are also nearly identical to the end-of-month no-

Table 7. Model-based Forecasts of Nominal CAD/USD and NOK/USD Exchange Rates

Method	Information	CAD	NOK
<i>Baseline information sets</i>			
BU-Recursive	Daily FX, oil	0.66 [0.000]	0.61 [0.000]
EoP NC	EoP FX	0.66 [0.000]	0.61 [0.000]
BUMIDAS	HF FX, oil	0.66 [0.000]	0.61 [0.000]
VAR	LF FX, oil	0.93 [0.000; 0.000]	0.92 [0.051; 0.000]
UMIDAS	LF FX, HF oil	0.96 [0.158; 0.000]	0.99 [0.435; 0.000]
RMIDAS	LF FX, HF oil	0.92 [0.000; 0.000]	0.90 [0.038; 0.000]
<i>High-frequency FX information</i>			
BUMIDAS-X	HF FX	0.66 [0.000; 0.479]	0.61 [0.000; 0.634]
UMIDAS-HF	HF FX, oil	0.77 [0.000; 0.005]	0.75 [0.000; 0.012]
RMIDAS-HF	HF FX, oil	0.79 [0.000; 0.037]	0.71 [0.000; 0.024]

Notes: MSFE ratios and first DM p -value relative to the monthly-average no-change; second DM p -value relative to BUMIDAS. LF/HF: low/high frequency.

change forecast, indicating that much of their gain comes from retaining the latest high-frequency observation on the exchange rate.

The final three specifications distinguish access to high-frequency target information from how that information is used. Removing oil from BUMIDAS has virtually no effect, suggesting little incremental predictive content from oil. Giving UMIDAS and RMIDAS high-frequency exchange-rate information produces much larger gains, lowering their MSFE ratios to 0.7–0.8. Nevertheless, both remain significantly less accurate than BUMIDAS. Thus, access to the high-frequency target is important, but exploiting that information parsimoniously provides an additional gain.

The relative gains are stable throughout the forecast sample and robust to alternative transformations, lag structures, and nominal effective exchange rates (Appendix A5.3). When exchange rates are instead end-of-month point sampled, the non-bottom-up methods become more accurate because they project directly from the latest observed exchange rate, avoiding the information loss from period averaging. Even then, none of the forecasts significantly outperform the end-of-month no-change forecast, and BUMIDAS significantly outperforms UMIDAS in all cases. Such robustness demonstrates that the gains from efficiently exploiting high-frequency target information are

substantially larger than the effects of conventional specification choices.

4.3 Headline Inflation

We next evaluate one-month-ahead forecasts of seasonally adjusted U.S. headline CPI inflation, measured in month-over-month percentage change. We target the 2026M8 vintage of CPI using mixed-frequency information from daily WTI oil and weekly retail gasoline prices, and the three-month Treasury bill rate. As a daily inflation proxy, we use the publicly available U.S. price index from the Billion Prices Project (BPP) over 2008M7–2015M8. The BPP index is constructed from online prices and has been shown to track official CPI and contain useful short-run inflation information (Cavallo and Rigobon, 2016). The short training sample provides a demanding case study; models use expanding window estimation, and out-of-sample forecasts cover 2012M1–2015M8. We also assess the effect of excluding BPP over the same sample and, separately, evaluate forecasts over a longer 2010–2025 sample using data beginning in 2000. High-frequency data are appropriately month-over-month transformed, and all mixed-frequency models use up to one month of high-frequency lags.

Given that the underlying DGP is unknown, the baseline BUMIDAS specification uses BIC to select the high-frequency terms (BUMIDAS-BIC), with AIC reported as robustness (BUMIDAS-AIC). To consider sparse extensions of BUMIDAS, we use coefficient-level Bayesian LASSO shrinkage (Bayesian LASSO) and LASSO with the penalty selected by BIC followed by post-selection fitting (Post-LASSO-BIC). The mixed-frequency benchmarks are UMIDAS and RMIDAS with Almon weights. Following Mogliani and Simoni (2021), we also evaluate Bayesian adaptive group-LASSO shrinkage applied to restricted Almon lag profiles (BMIDAS-AGL).

As monthly benchmarks, we report an unobserved-components model (UCM) following Stock and Watson (2007), a random walk with drift (RWWD), and a Bayesian VAR with 12 lags and

conjugate Minnesota-type shrinkage in the spirit of Carriero et al. (2015) (BVAR-CCM). The BVAR includes monthly CPI inflation, oil and gasoline price growth, and interest-rate changes.

Table 8. Model-based Forecasts of U.S. Headline Inflation

Method	2012M2–2015M8 with BPP	2012M2–2015M8 without BPP	2010–2025 without BPP
<i>Monthly benchmarks</i>			
RWWD	0.78 (0.185)	0.78 (0.185)	1.07 (0.629)
UCM	0.99 (0.373) [0.014]	0.99 (0.373) [0.020]	0.91 (0.270) [0.062]
BVAR-CCM	0.73 (0.060) [0.029]	0.73 (0.060) [0.029]	0.84 (0.077) [0.065]
<i>BUMIDAS and sparse unrestricted models</i>			
BUMIDAS-BIC	0.54 (0.024)	0.54 (0.024)	0.71 (0.042)
BUMIDAS-AIC	0.63 (0.036) [0.106]	0.57 (0.043) [0.318]	0.73 (0.053) [0.099]
Bayesian LASSO	0.55 (0.016) [0.528]	0.54 (0.022) [0.476]	0.73 (0.029) [0.353]
Post-LASSO-BIC	0.66 (0.030) [0.080]	0.54 (0.019) [0.484]	0.75 (0.039) [0.271]
<i>MIDAS benchmarks</i>			
UMIDAS	3.42 (1.000) [0.000]	9.18 (0.987) [0.009]	0.85 (0.149) [0.112]
RMIDAS	0.66 (0.095) [0.228]	0.58 (0.042) [0.335]	0.70 (0.036) [0.711]
BMIDAS-AGL	0.82 (0.157) [0.014]	0.65 (0.028) [0.090]	0.68 (0.025) [0.807]

Notes: Monthly benchmarks use monthly data; all other models use mixed-frequency monthly and daily data. Entries are MSFE ratios relative to the no-change forecast. Parentheses report one-sided p -values relative to no change; brackets report one-sided DM p -values relative to BUMIDAS-BIC.

Table 8 shows that BUMIDAS-BIC and Bayesian LASSO perform best in the short sample, with BUMIDAS-AIC, Post-LASSO-BIC, and RMIDAS also improving significantly at the 10 percent level on the no-change forecast. In the short sample, UMIDAS lags are truncated to those estimable, which explains its poor performance with and without BPP. Excluding BPP leaves the MSFE ratio for BUMIDAS-BIC unchanged and improves all other mixed-frequency methods except UMIDAS. Taken together, the results indicate little incremental forecast benefit from BPP in this short sample, conditional on the other high-frequency predictors.

Over the longer 2010–2025 evaluation with BPP excluded, differences across the mixed-frequency methods narrow considerably. BMIDAS-AGL has the lowest MSFE ratio, followed by RMIDAS and BUMIDAS-BIC; neither BMIDAS-AGL nor RMIDAS differs significantly from BUMIDAS-BIC. UMIDAS also improves, with its MSFE ratio falling to 0.85. Overall, the application confirms

that high-frequency information provides substantial gains relative to monthly benchmarks and that RMIDAS performs well when informative high-frequency predictors are available. Moreover, coefficient-level sparsity did well, but did not systematically improve on BUMIDAS-BIC, whereas group-level shrinkage became competitive in the larger sample.

4.4 GDP Nowcasting

We next consider textbook real-time nowcasts of quarterly U.S. real GDP growth. Building on Clements and Galvão (2008), we employ strict real-time vintages of quarterly GDP and four monthly indicators typically available with a one-month publication lag: industrial production, housing starts, nonfarm payroll employment, and capacity utilization. Models are estimated using an expanding window on data beginning in 1985 for a real-time evaluation sample of 2000Q1–2025Q2 and target the 2026Q2 vintage. Three nowcasts are constructed using the available information for each month of the quarter.

We compare bridge equations, BUMIDAS-BIC, UMIDAS, and RMIDAS, building upon existing conventional practice (Clements and Galvão, 2009; Foroni et al., 2015a; Schumacher, 2016). UMIDAS and RMIDAS use the six most recent monthly observations of each indicator. While UMIDAS applications commonly employ BIC, the reductions are motivated by quarterly blocks, whereas BUMIDAS-BIC selects individual unrestricted terms using BIC. Every approach is repeated when all four indicators are projected on jointly (combined models) or by equally weighting forecasts from four single-indicator models (pooled models). An AR(1) provides the low-frequency benchmark and is the best-performing autoregressive alternative; see Appendix A5.5.

Table 9 shows substantial gains from incorporating within-quarter monthly information. At the first monthly vintage, BUMIDAS-BIC is the most accurate in both the combined and pooled comparisons, reducing MSFE by 46 and 31 percent relative to the AR(1). As additional information

Table 9. Model-based Real-Time Nowcasts of U.S. Real GDP Growth

	Bridge	BUMIDAS-BIC	UMIDAS	RMIDAS
<i>Panel A. Combined models</i>				
Month 1	1.10 (0.673)	0.54 (0.133)	0.83 (0.191)	1.35 (0.760)
Month 2	0.25 (0.151)	0.52 (0.181)	0.37 (0.159)	0.55 (0.185)
Month 3	0.11 (0.138)	0.23 (0.149)	0.16 (0.146)	0.11 (0.140)
<i>Panel B. Pooled models</i>				
Month 1	0.87 (0.111)	0.69 (0.127)	0.94 (0.369)	1.31 (0.769)
Month 2	0.15 (0.140)	0.19 (0.143)	0.13 (0.142)	0.23 (0.140)
Month 3	0.18 (0.137)	0.34 (0.148)	0.12 (0.140)	0.16 (0.141)

Notes: MSFE ratios are relative to the recursive AR(1); one-sided Diebold–Mariano p -values are in parentheses. Month 1–3 denote the three within-quarter information sets. Pooled forecasts equally weight the four single-indicator models. Bold indicates the lowest MSFE by forecast origin.

arrives, forecast gains become broad-based, with Bridge and UMIDAS particularly competitive. Compared with combined models, pooling lowers MSFE for all four methods in the second month but only for UMIDAS in the third. Although the MSFE reductions are economically large, the DM tests do not reject equal predictive accuracy relative to the AR(1) at the 10 percent level.

5 Conclusion

This paper uses the bottom-up approach to derive the optimal direct mixed-frequency representation for temporally aggregated VARMA(p, q) processes. When the relevant high-frequency information is observed or recoverable, the most recent p observations and q innovations suffice. Temporal aggregation changes the coefficients but not the number of required high-frequency terms. BUMIDAS yields the same population forecast as recursive bottom-up forecasting, permitting parsimonious unrestricted estimation even for daily-to-monthly and daily-to-quarterly aggregation.

Model specification cannot eliminate losses from high-frequency information that the available data do not reveal. When the high-frequency target is observed, information criteria can recover the parsimonious BUMIDAS structures. When the target is proxied or unavailable, the model

approximates a filter that infers the latent high-frequency state from available high-frequency and aggregate lags. A proxy’s incremental predictive information must be weighed against the cost of estimating a larger specification.

BUMIDAS delivers substantial gains for persistent processes under daily-to-monthly and daily-to-quarterly aggregation. Much of the gain comes from retaining the target’s high-frequency information, while parsimony further improves finite-sample accuracy. Simulations and applications show improvements over conventional specifications that are often statistically significant, particularly against UMIDAS. Under incomplete information, its parsimonious approximations also outperform conventional MIDAS in monthly and quarterly simulations.

The practical implication is to choose mixed-frequency specifications according to the available information and the underlying dynamics. More generally, bottom-up modelling can guide extensions to larger systems, using selection and shrinkage to balance information gains against estimation costs, and to nonlinear or time-varying dynamics, including structural breaks. The bottom-up approach therefore provides both an efficiency benchmark and a framework for developing mixed-frequency models grounded in the underlying high-frequency dynamics.

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Online Appendix

A1 Data Appendix

All empirical series are publicly available. Sources and series identifiers are listed below.

For financial data, let $S_{t,i}$ denote the value on trading day $i = 1, \dots, n$ of month t . The period-average level is the simple average of daily values, $\bar{S}_t = \frac{1}{n} \sum_{i=1}^n S_{t,i}$, while the end-of-month value is $S_{t,n}$.

U.S. 3-Month Treasury Constant-Maturity Rate

Daily secondary-market yields (percent p.a., not seasonally adjusted) are taken from the Federal Reserve Bank of St. Louis *FRED* database, series **DGS3MO** (“Market Yield on U.S. Treasury Securities at 3-Month Constant Maturity, Quoted on an Investment Basis”).

Exchange Rates

Daily nominal exchange rates are taken from the Federal Reserve Bank of St. Louis *FRED* database: series DEXCAUS (“Canadian Dollars to U.S. Dollar Spot Exchange Rate”) and DEXNOUS (“Norwegian Kroner to U.S. Dollar Spot Exchange Rate”). Both series are reported as units of domestic currency per U.S. dollar and are based on the Federal Reserve Board’s H.10 daily noon buying rates in New York City. The monthly-average exchange rate is the simple average of daily observations within each month, while the end-of-month rate is the final daily observation.

Daily narrow nominal effective exchange rate (NEER) indices for Canada and Norway (index, 2020 = 100, not seasonally adjusted) are obtained from the Bank for International Settlements (BIS) **Effective Exchange Rates** database, series keys D.N.N.CA and D.N.N.NO in the **WS_EER** data set. The indices are geometric trade-weighted averages of bilateral exchange rates based on the BIS narrow basket, with time-varying weights derived from manufacturing trade and accounting

for third-market competition.

Consumer Price Index (CPI)

Real-time headline CPI (seasonally adjusted, index 2015 = 100) is obtained from the Federal Reserve Bank of Philadelphia’s Real-Time Data Research Center, vintage series **PCPI**. (Vintages “PCPI98M11–PCPI26M8”). The Bureau of Labor Statistics publishes the CPI with a one-month delay that is available around the middle of the subsequent month. Forecasts target the PCPI26M8 vintage.

West Texas Intermediate Crude Oil

Daily WTI spot prices (US\$ barrel) come from the U.S. Energy Information Administration’s (EIA) *Petroleum and Other Liquids* portal, series **RWTC** (“Cushing, OK WTI Spot Price FOB (Dollars per Barrel)”). The *EIA* posts the series each business day with no subsequent revision.

U.S. Retail Gasoline Prices

Weekly retail gasoline prices are taken from Table 14 of the *EIA Weekly Petroleum Status Report* **Sourcekey: EMM_EPMRU_PTE_NUS_DPG**. We used the series “U.S. Regular Weekly Conventional Retail Gasoline Prices (Dollars per Gallon)” for which the longest history is available. The EIA’s weekly price corresponds to the price on the Monday of the week and is published on the same day, except on government holidays, for which the data is released on Tuesday, see **Definitions, Sources and Explanatory Notes**. This price is observable in real time.

Real-time GDP data and monthly predictors

All data for the GDP nowcasting exercise are obtained from the Federal Reserve Bank of Philadelphia’s Real-Time Data Set for Macroeconomists, using the **Full-Time Series History files**. Each

monthly vintage records the historical observations available at its reference date, preserving publication lags and subsequent revisions. We use real GDP (ROUTPUT), total industrial production (IPT), housing starts (HSTARTS), nonfarm payroll employment (EMPLOY), and total capacity utilization (CUT). The source data are reported in levels; quarterly log differences are used for GDP growth and the monthly predictor growth rates used in the forecasting models.

A2 Proofs and Technical Details

Time indices and dynamics

Symbol	Meaning
t, T	Low-frequency period index and last observed period at the forecast origin.
i, n	Within-period position and observations per period.
h	Forecast horizon in low-frequency periods.
L	High-frequency lag operator: $LZ_{t,i} = Z_{t,i-1}$, with $Z_{t,0} = Z_{t-1,n}$.
$Z_{t,i}$	High-frequency vector; in the bivariate theory, $(Y_{t,i}, X_{t,i})'$ contains the target and predictor.
$\bar{Z}_t$	Period average: $\bar{Z}_t = n^{-1} \sum_{i=1}^n Z_{t,i}$.
p, q	Autoregressive and moving-average orders of the high-frequency VARMA process.

A2.1 Proof of Proposition 1

Proof. The daily VARMA process is given by:

$$Y_{t,i} = \sum_{j=1}^p A_j Y_{t,i-j} + \varepsilon_{t,i} + \sum_{\ell=1}^q \Theta_\ell \varepsilon_{t,i-\ell} \quad (12)$$

At horizon $T+1, 1$, the optimal forecast is:

$$\mathbb{E}_{T,n}[Y_{T+1,1}] = \sum_{j=1}^p A_j Y_{T,n-j+1} + \sum_{\ell=1}^q \Theta_\ell \varepsilon_{T,n-\ell+1} \quad (13)$$

By forward substitution, we get at horizon $T + h, i$:

$$\mathbb{E}_{T,n}[Y_{T+h,i}] = \sum_{j=1}^p \Phi_j(h, i) Y_{T,n-j+1} + \sum_{\ell=1}^q \Gamma_{\ell}(h, i) \varepsilon_{T,n-\ell+1}, \quad (14)$$

where $\Phi_j(h, i)$ are the coefficients for the VAR terms and $\Gamma_{\ell}(h, i)$ are the coefficients for the VMA terms. In the first p forecasts, the autoregressive coefficients are generated by:

$$\Phi_j(1, i) = \begin{cases} A_{j+i-1} + \sum_{1 \leq s \leq i-1} A_s \Phi_j(1, i-s) & \text{for } j+i-1 \leq p \\ \sum_{1 \leq s \leq i-1} A_s \Phi_j(1, i-s) & \text{otherwise} \end{cases} \quad (15)$$

with initial conditions $\Phi_j(1, 1) = A_j$ for $j = 1, \dots, p$ and $A_s = 0$ for $s > p$. In the first q forecasts, the moving average coefficients are generated by:

$$\Gamma_{\ell}(1, i) = \begin{cases} \Theta_{\ell+i-1} + \sum_{1 \leq s \leq i-1} A_s \Gamma_{\ell}(1, i-s) & \text{for } \ell+i-1 \leq q \\ \sum_{1 \leq s \leq i-1} A_s \Gamma_{\ell}(1, i-s) & \text{otherwise} \end{cases} \quad (16)$$

with initial conditions $\Gamma_{\ell}(1, 1) = \Theta_{\ell}$ for $\ell = 1, \dots, q$. Thereafter, the coefficients evolve according to the recurrence relations:

$$\begin{aligned} \Phi_j(h, i) &= A_1 \Phi_j(h, i-1) + \dots + A_p \Phi_j(h, i-p), \\ \Gamma_{\ell}(h, i) &= A_1 \Gamma_{\ell}(h, i-1) + \dots + A_p \Gamma_{\ell}(h, i-p), \end{aligned} \quad (17)$$

For $h \geq 2$, the recursion continues across low-frequency periods according to $\Phi_j(h, r) = \Phi_j(h-1, n+r)$ and $\Gamma_{\ell}(h, r) = \Gamma_{\ell}(h-1, n+r)$ for $r \leq 0$, so that lags extending beyond the beginning of period h are taken from the end of period $h-1$.

Thus, the autoregressive terms $\Phi_j(h, i)$ are functions of only the autoregressive parameters

$A_1, \dots, A_p$, but the moving average terms $\Gamma_\ell(h, i)$ are functions of both autoregressive parameters $A_1, \dots, A_p$ and the moving average parameters $\Theta_1, \dots, \Theta_q$. The expected trajectory of the process $Y_{T+h,i}$ conditional on T, n is then given by:

$$\mathbb{E}_{T,n} [Y_{T+h,i}] = \sum_{j=1}^p \Phi_j(h, i) Y_{T,n-j+1} + \sum_{\ell=1}^q \Gamma_\ell(h, i) \varepsilon_{T,n-\ell+1}. \quad (18)$$

Then, we can get the forecast of $\bar{Y}_{T+h}$ by aggregation:

$$\mathbb{E}_{T,n} [\bar{Y}_{T+h}] = \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^p \Phi_j(h, i) Y_{T,n-j+1} + \frac{1}{n} \sum_{i=1}^n \sum_{\ell=1}^q \Gamma_\ell(h, i) \varepsilon_{T,n-\ell+1}. \quad (19)$$

Hence, we have $\beta_{h,j} = \frac{1}{n} \sum_{i=1}^n \Phi_j(h, i)$ and $\alpha_{h,\ell} = \frac{1}{n} \sum_{i=1}^n \Gamma_\ell(h, i)$ □

A2.2 Proof of Proposition 2

Proof. The MSFE is defined as:

$$\begin{aligned} & MSFE(\bar{Z}_{t+h} | \mathcal{F}_{T,n}) \\ &= \mathbb{E}[(\bar{Z}_{T+h} - \hat{\bar{Z}}_{T+h}(\mathcal{F}_{T,n}))(\bar{Z}_{T+h} - \hat{\bar{Z}}_{T+h}(\mathcal{F}_{T,n}))' | \mathcal{F}_{T,n}] \\ &= \mathbb{V}[\bar{Z}_{T+h} | \mathcal{F}_{T,n}] + [\mathbb{E}(\bar{Z}_{T+h} | \mathcal{F}_{T,n}) - \hat{\bar{Z}}_{T+h}(\mathcal{F}_{T,n})][\mathbb{E}(\bar{Z}_{T+h} | \mathcal{F}_{T,n}) - \hat{\bar{Z}}_{T+h}(\mathcal{F}_{T,n})]', \\ &= \mathbb{V}[\bar{Z}_{T+h} | \mathcal{F}_{T,n}] + B(\bar{Z}_{T+h} | \mathcal{F}_{T,n}), \end{aligned} \quad (20)$$

where conditional MSFE equals the conditional variance plus the squared deviation of the forecast from the conditional mean. The variance term can further be decomposed by the Law of Total

Variance to obtain:

$$\begin{aligned}
\mathbb{V}[\bar{Z}_{T+h}|\mathcal{F}_{T,n}] &= \mathbb{E}[\mathbb{V}(\bar{Z}_{T+h}|Z_{t,n}, \mathcal{F}_{T,n})|\mathcal{F}_{T,n}] + \mathbb{V}[\mathbb{E}(\bar{Z}_{T+h}|Z_{t,n}, \mathcal{F}_{T,n})|\mathcal{F}_{T,n}] \\
&= \mathbb{E}[\mathbb{V}(\bar{Z}_{T+h}|Z_{t,n})|\mathcal{F}_{T,n}] + \mathbb{V}[\mathbb{E}(\bar{Z}_{T+h}|Z_{t,n})|\mathcal{F}_{T,n}] \\
&= \mathbb{V}_{T,n}^{BUMIDAS}[\bar{Z}_{T+h}] + \mathbb{V}[\beta_h Z_{T,n}|\mathcal{F}_{T,n}] \\
&= \mathbb{V}_{T,n}^{BUMIDAS}[\bar{Z}_{T+h}] + \beta_h \mathbb{V}[Z_{T,n}|\mathcal{F}_{T,n}] \beta_h'
\end{aligned}$$

Taking expectations with respect to $\mathcal{F}_{T,n}$ on both sides yields (6). □

A2.3 State Space Representation of Proxy Case

The state is a $(2n \times 1)$ stacked vector of the N high-frequency observations (the total number of days in an aggregate period) for both Y and X .

$$\tilde{Z}_T = \begin{pmatrix} Z_{T,1} \\ Z_{T,2} \\ \vdots \\ Z_{T,n} \end{pmatrix}_{2n \times 1} = \begin{pmatrix} Y_{T,1} \\ X_{T,1} \\ \vdots \\ Y_{T,n} \\ X_{T,n} \end{pmatrix} \quad (21)$$

The measurement is a $(2n+1) \times 1$ vector of the new observables that arrives at every low-frequency period T . They include the aggregate $\bar{Y}_T$, the n high-frequency observations of X and the n

high-frequency observations of the proxy variable Y^* .

$$W_T = \begin{pmatrix} \bar{Y}_T \\ X_{T,1} \\ \vdots \\ X_{T,n} \\ Y_{T,1}^* \\ \vdots \\ Y_{T,n}^* \end{pmatrix} \quad (22)$$

The state innovation $\tilde{u}_T$ is a linear combination of daily innovations in the current month. In particular, for a day T, i we have:

$$u_{T,i} \equiv \sum_{j=1}^i A^{i-j} \varepsilon_{T,j}. \quad (23)$$

Since $\varepsilon_{T,i}$ is a (2×1) bivariate innovation, $u_{t,i} = (u_{T,i}^Y, u_{T,i}^X)$ is also (2×1) . Thus we have a $(2n \times 1)$ stacked vector of state innovations.

$$\tilde{u}_T = \begin{pmatrix} u_{T,1} \\ \vdots \\ u_{T,n} \end{pmatrix}. \quad (24)$$

The measurement innovations consists of three blocks. The first two are the measurement “errors” for $\bar{Y}$ and the daily X ’s, which are observed by the practitioner. Therefore, their errors are all 0. The last block is the n noise terms for the daily proxies:

$$\tilde{\nu}_T = \begin{pmatrix} 0 \\ 0 \\ \nu_T \end{pmatrix}, \quad \nu_T = \begin{pmatrix} \nu_{T,1} \\ \vdots \\ \nu_{T,n} \end{pmatrix}. \quad (25)$$

Putting these vectors together in state space representation we get:

$$\underset{2N \times 1}{\tilde{Z}_T} = \underset{2n \times 1}{\mathcal{A}} \underset{2n \times 1}{\tilde{Z}_{T-1}} + \underset{2n \times 1}{\tilde{u}_T}, \text{ (State Equation)}$$

$$\underset{(2n+1) \times 1}{W_T} = \underset{2n \times 1}{C} \underset{2n \times 1}{\tilde{Z}_T} + \underset{(2n+1) \times 1}{\tilde{\nu}_T}. \text{ (Measurement Equation)}$$

Finally, we organize the coefficient matrix $\mathcal{A}$ and measurement matrix C so that the dimensions of the system agree. $\mathcal{A}$ is of dimension $(2n \times 2n)$ given by:

$$\mathcal{A} = \begin{pmatrix} 0 & 0 & \cdots & 0 & A \\ 0 & 0 & \cdots & 0 & A^2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & A^N \end{pmatrix}. \quad (26)$$

And C is of dimension $((2n+1) \times 2n)$. Again, it is easier to understand this matrix if it is partitioned into the aggregate, X and proxy blocks:

$$\underset{(2n+1) \times 2n}{C} = \begin{pmatrix} \underset{1 \times 2n}{C_{\bar{Y}}} \\ \underset{N \times 2n}{C_X} \\ \underset{N \times 2n}{C_{Y^*}} \end{pmatrix}, \quad (27)$$

where:

$$\underset{1 \times 2n}{C_{\bar{Y}}} = \begin{pmatrix} \frac{1}{n} & 0 & \frac{1}{n} & 0 & \cdots & \frac{1}{n} & 0 \end{pmatrix}. \quad (28)$$

$$C_X = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & & & & \ddots & & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix}, \quad (29)$$

$$C_{Y^*} = \begin{pmatrix} B & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & B & 0 & \cdots & 0 & 0 \\ \vdots & & & & \ddots & & \vdots \\ 0 & 0 & 0 & 0 & \cdots & B & 0 \end{pmatrix}. \quad (30)$$

A2.4 Infinite Representation of Filtered State

The filtering of the current state is given by:

$$\hat{\hat{Z}}_{T|T} = \hat{\hat{Z}}_{T|T-1} + K_T(W_T - C\hat{\hat{Z}}_{T|T-1}), \quad (31)$$

where K_T denotes the Kalman gain at time T . To obtain the infinite representation, we recursively apply the filtering equation:

$$\begin{aligned} \hat{\hat{Z}}_{T|T} &= \hat{\hat{Z}}_{T|T-1} + K_T(W_T - C\hat{\hat{Z}}_{T|T-1}) \\ &= (I - K_TC)\mathcal{A}\hat{\hat{Z}}_{T-1|T-1} + K_TW_T \\ &= (I - K_TC)\mathcal{A}((I - K_{T-1}C)\mathcal{A}\hat{\hat{Z}}_{T-2|T-2} + K_{T-1}W_{T-1}) + K_TW_T \\ &= (I - K_TC)\mathcal{A}(I - K_{T-1}C)\mathcal{A}\hat{\hat{Z}}_{T-2|T-2} + (I - K_TC)\mathcal{A}K_{T-1}W_{T-1} + K_TW_T \\ &= \sum_{j=0}^{\infty} \left\{ \left[\prod_{k=0}^{j-1} (I - K_{T-k}C)\mathcal{A} \right] K_{T-j} \right\} W_{T-j}. \end{aligned} \quad (32)$$

Then, note that $\hat{\tilde{Z}}_{T+h|T} = \mathcal{A}^h \hat{\tilde{Z}}_{T|T}$. Thus, we define $\Theta_{T,j} = \mathcal{A}^h \left[\prod_{k=0}^{j-1} (I - K_{T-k}C) \mathcal{A} \right] K_{T-j}$ to obtain the representation:

$$\hat{\tilde{Z}}_{T+h|T} = \sum_{j=0}^{\infty} \Theta_{T,j} W_{T-j}.$$

A2.5 Proof of Proposition 3

The weights $\Theta_{T,j} = \mathcal{A}^h \left[\prod_{k=0}^{j-1} (I - K_{t-k}C) \mathcal{A} \right] K_{T-j}$ depend on Kalman gain:

$$K_T = \Sigma_{T|T-1} C' (C \Sigma_{T|T-1} C' + \mathbb{V}(\tilde{\nu}_T))^{-1}, \quad (33)$$

so we just have to prove that the gain for proxies is 0 for any T . Observe that K_t is of dimension $(2n \times (2n + 1))$ since:

$$K_T = \Sigma_{T|T-1} \begin{matrix} C' \\ 2n \times (2n+1) \end{matrix} \left(\begin{matrix} C & \Sigma_{T|T-1} C' \\ (2n+1) \times 2n & 2n \times 2n \end{matrix} + \begin{matrix} \mathbb{V}(\tilde{\nu}_T) \\ (2n+1) \times (2n+1) \end{matrix} \right)^{-1}. \quad (34)$$

We partition it into two blocks: a vector of non-proxies (i.e. $\bar{Y}$ and X) and proxies (i.e. Y^*). For ease of notation, we write $\Sigma_{T|T-1} = \Sigma$ for this proof. Then we have:

$$\begin{pmatrix} K_{\bar{Y},X}^U & K_T^{Y^*} \end{pmatrix} = \begin{pmatrix} \Sigma C'_{\bar{Y},X} & \Sigma C'_{Y^*} \end{pmatrix} \left[\begin{pmatrix} C_{\bar{Y},X} \Sigma C'_{\bar{Y},X} & C_{\bar{Y},X} \Sigma C'_{Y^*} \\ C_{Y^*} \Sigma C'_{\bar{Y},X} & C_{Y^*} \Sigma C'_{Y^*} + \Sigma_{\nu} \end{pmatrix} \right]^{-1}. \quad (35)$$

Using a Schur complement we get:

$$\begin{aligned} K_T^{Y^*} &= \left[\Sigma - \Sigma C'_{\bar{Y},X} \left(C_{\bar{Y},X} \Sigma C'_{\bar{Y},X} \right)^{-1} C_{\bar{Y},X} \Sigma \right] C'_{Y^*} \\ &\quad \times \left[C_{Y^*} \Sigma C'_{Y^*} + \Sigma_{\nu} - C_{Y^*} \Sigma C'_{\bar{Y},X} \left(C_{\bar{Y},X} \Sigma C'_{\bar{Y},X} \right)^{-1} C_{\bar{Y},X} \Sigma C'_{Y^*} \right]^{-1}. \end{aligned} \quad (36)$$

As shown in Appendix A2.3, C_{Y^*} is a matrix of 0's and proxy strength B . Therefore, if $B = 0$, then C_{Y^*} is a matrix of 0's and immediately we get $K_T^{Y^*} = 0$. Similarly, as $\sigma_\nu^2 \rightarrow \infty$, the proxy-measurement block of Σ_ν diverges, causing the corresponding inverse term to converge to zero. Hence, $K_T^{Y^*} \rightarrow 0$.

To prove the second point, note that the measurement matrices in A2.3, $C_{\bar{Y}}$ and C_{Y^*} , are related as follows:

$$C_{\bar{Y}} = \frac{1}{Bn} \mathbf{1}_n' C_{Y^*}.$$

Hence, when the proxy noise is zero, the aggregate measurement is a linear combination of the high-frequency proxy measurements and is therefore informationally redundant. In this case,

$$C\Sigma C' + \mathbb{V}(\tilde{\nu}_T) \tag{37}$$

reduces to $C\Sigma C'$, which is singular if the redundant aggregate row is retained. Removing the aggregate row yields the equivalent reduced measurement matrix

$$\tilde{C}_{2n \times 2n} = \begin{pmatrix} C_X \\ C_{Y^*} \end{pmatrix}. \tag{38}$$

Then, the Kalman gain is now given by:

$$K_t = \Sigma \tilde{C}' (\tilde{C} \Sigma \tilde{C}')^{-1} = \Sigma \tilde{C}' \tilde{C}'^{-1} \Sigma^{-1} \tilde{C}^{-1} = \tilde{C}^{-1}. \tag{39}$$

Thus, $K_T \tilde{C} = \begin{smallmatrix} I \\ 0 \end{smallmatrix}_{2n \times 2n}$, and $(I - K_T \tilde{C}) = 0$. It follows that $\Theta_{T,j} = 0$, $j \geq 1$, so the filtered state depends only on contemporaneous high-frequency information. Moreover, since $\mathcal{A}$ has only its final block column nonzero, the h -step forecast depends on the current state only through $Z_{T,n}$. Hence,

the resulting forecast coincides with the BUMIDAS specification.

Finally, note that $\Sigma_{T|T} = \mathbb{V}[\tilde{Z}_T|\mathcal{F}_T] = (I - K_T\tilde{C})\Sigma_{T|T-1}$. Hence, when the proxy is perfect, $\Sigma_{T|T} = 0$, as required.

A3 Special Cases of Proposition 1

A3.1 Vector Autoregressions

Corollary 2. The optimal mixed-frequency direct forecast of a temporally aggregated VAR(p) with $n \geq 2$ and forecast horizon h is given by

$$\mathbb{E}_{T,n} [\bar{Z}_{T+h}] = \sum_{j=1}^p \beta_{h,j} Z_{T,n-j+1}, \quad (40)$$

where $\beta_{h,j}$ are functions of the autoregressive matrices $A_1, \dots, A_p$.

A3.2 Vector Moving Averages

Corollary 3. The optimal direct forecast of a temporally aggregated (V)MA(q) with $n \geq 2$ and forecast horizon h is given by:

$$\mathbb{E}_{T,n} [\bar{Y}_{T+h}] = \begin{cases} \sum_{\ell=1}^q \alpha_{h,\ell} \varepsilon_{T,n-\ell+1}, & \text{if } (h-1)n < q \\ 0, & \text{otherwise.} \end{cases}$$

For a (V)MA(q), past innovations affect forecasts at most q high-frequency steps ahead. The period-average forecast is therefore zero once its first observation lies beyond this horizon, that is, when $(h-1)n \geq q$. Temporal aggregation changes the weights on past innovations but does not extend their predictive horizon.

The intuition of this result is straightforward: a finite order MA process carries no serial depen-

dence beyond q daily shocks, so once the forecast horizon exceeds those q steps, the conditional mean of every remaining shock is zero. More importantly, this memoryless property is inherited from the underlying daily innovations rather than the lower-frequency moving average. This suggests that aggregation alone does not reintroduce memory.

A3.3 Point-in-Time Sampling

Aside from temporally aggregated data, one may also be interested in point-in-time sampling, including the use of end-of-period data. MIDAS applications commonly model lower-frequency end-of-period returns (e.g., monthly) using high-frequency predictors (e.g., daily or hourly observations). We show that the principles of BUMIDAS can be readily applied in this setting.

Corollary 4. The bottom-up mixed-frequency direct forecast of a point-in-time sampled VARMA(p, q) at point v and forecast horizon h is given by

$$\mathbb{E}_{T,n} [Y_{T+h,v}] = \sum_{j=1}^p \beta_{h,j} Y_{T,n-j+1} + \sum_{\ell=1}^q \alpha_{h,\ell} \varepsilon_{T,n-\ell+1}$$

Proof. The expected trajectory of the process $Y_{T+h,i}$ conditional on T, n is given by equation 18.

Then, we can get the forecast of $Y_{T+h,v}$ by sampling the forecast at only point v :

$$\mathbb{E}_{T,n} [Y_{T+h,v}] = \sum_{j=1}^p \Phi_j(h, v) Y_{T,n-j+1} + \sum_{\ell=1}^q \Gamma_\ell(h, v) \varepsilon_{T,n-\ell+1}. \quad (41)$$

Hence, we have $\beta_{h,j} = \Phi_j(h, v)$ and $\alpha_{h,\ell} = \Gamma_\ell(h, v)$ as required. $\square$

The implication is that regardless of modeling temporally aggregated or point-in-time sampled VARMA(p, q) processes, the structure depends entirely on the degree of p and q , but not the time sampling n .

A3.4 Nowcasting

Consider the high-frequency VARMA(p, q) process $a(L)Z_{t,i} = b(L)\varepsilon_{t,i}$ and the period average $\bar{Z}_t = n^{-1} \sum_{i=1}^n Z_{t,i}$, as in Section 2.1. Suppose all relevant high-frequency observations, including the target, are available through day $m \in \{1, \dots, n-1\}$ of period t , and the required innovations are observed or recoverable. Let $\mathbb{E}_{t,m}$ denote expectation conditional on this information. The bottom-up nowcast combines the observed contribution to the period average with forecasts of the remaining observations.

Corollary 5 (BUMIDAS Nowcasting). The optimal nowcast of the temporally aggregated VARMA(p, q) is

$$\mathbb{E}_{t,m}[\bar{Z}_t] = \frac{1}{n} \sum_{i=1}^m Z_{t,i} + \sum_{j=1}^p \beta_{m,j} Z_{t,m-j+1} + \sum_{\ell=1}^q \alpha_{m,\ell} \varepsilon_{t,m-\ell+1}, \quad (42)$$

where the coefficient matrices depend on the underlying dynamics, the aggregation frequency n , and the within-period origin m .

Proof. Since the first m observations are known,

$$\mathbb{E}_{t,m}[\bar{Z}_t] = \frac{1}{n} \sum_{i=1}^m Z_{t,i} + \frac{1}{n} \sum_{r=1}^{n-m} \mathbb{E}_{t,m}[Z_{t,m+r}].$$

Forward substitution expresses each remaining daily forecast as a linear function of the latest p high-frequency observations and q innovations. If $\Phi_j(r)$ and $\Gamma_\ell(r)$ denote their respective r -step forecast coefficients, then

$$\beta_{m,j} = \frac{1}{n} \sum_{r=1}^{n-m} \Phi_j(r), \quad \alpha_{m,\ell} = \frac{1}{n} \sum_{r=1}^{n-m} \Gamma_\ell(r),$$

which gives (42). □

Thus, at each within-period origin, the number of estimated coefficients remains determined by p and q ; the origin changes the coefficient values and the dates of the conditioning observations. The observed contribution enters with its known aggregation weights. In practice, a direct BUMIDAS nowcasting regression would subtract the observed contribution from the dependent variable and project the remainder on the latest high-frequency state.

Illustrative Example: Daily AR(1)

Suppose

$$y_{t,i} = \rho y_{t,i-1} + \varepsilon_{t,i}, \quad i = 1, \dots, n,$$

with conditionally mean-zero innovations. For $n = 2$, observing $y_{t,1}$ halfway through the period gives

$$\mathbb{E}_{t,1}[\bar{y}_t] = \frac{1}{2} (y_{t,1} + \mathbb{E}_{t,1}[y_{t,1+r}]) = \frac{1+\rho}{2} y_{t,1},$$

where $r = 1$. The direct BUMIDAS nowcasting specification is therefore

$$\bar{y}_t = \beta_1 y_{t,1} + u_{t,1}, \quad \beta_1 = \frac{1+\rho}{2}, \quad u_{t,1} = \frac{\varepsilon_{t,2}}{2}.$$

The single regressor captures both the observed contribution and the information needed to forecast the remaining day.

For general n , after observing the first m days,

$$\mathbb{E}_{t,m}[\bar{y}_t] = \frac{1}{n} \sum_{i=1}^m y_{t,i} + \left(\frac{1}{n} \sum_{h=1}^{n-m} \rho^h \right) y_{t,m}. \quad (43)$$

The latest observation suffices to forecast the unobserved remainder. Earlier observations within the current period enter through their known contribution to the aggregate, requiring no additional estimated coefficients.

A4 Additional Simulations

Section A4.2 provides additional results on model specification using statistical tests and information criteria when the underlying high-frequency state is observed. Section A4.3 reports corresponding results when the high-frequency target is proxied or unobserved. Sections A4.4 and A4.5 provide additional forecast results for the univariate and multivariate DGPs, respectively.

A4.1 Simulation Data-Generating Processes

Let $Z_{t,i} = (y_{1,t,i}, y_{2,t,i})'$ denote the daily state on day $i = 1, \dots, n$ of low-frequency period t , consistent with the notation in Section 2. Lags extend across low-frequency periods, with $Z_{t,0} = Z_{t-1,n}$. The structural innovations satisfy

$$\varepsilon_{t,i} = (\varepsilon_{1,t,i}, \varepsilon_{2,t,i})' \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_2)$$

and enter through Λ

$$\Lambda = \begin{bmatrix} 1 & \eta_1 \\ \eta_2 & 1 \end{bmatrix}.$$

The VAR(1) DGP, corresponding to $p = 1$ and $q = 0$,

$$Z_{t,i} = A_1 Z_{t,i-1} + \Lambda \varepsilon_{t,i}. \tag{44}$$

The baseline triangular calibration sets

$$A_1^{\text{tri}} = \begin{bmatrix} 0.99 & 0 \\ 0.50 & 0.99 \end{bmatrix},$$

with $\eta_1 = 0$ and $\eta_2 = 0.5$. Thus, $y_{1,t,i}$ depends only on its own lag and provides the univariate

target used in the specification exercises, whereas $y_{2,t,i}$ is the multivariate target.

The feedback calibration sets

$$A_1^{\text{feed}} = \begin{bmatrix} 0.90 & 0.10 \\ 0.095 & 0.90 \end{bmatrix},$$

with $\eta_1 = 0$ and $\eta_2 = 0$, so that each variable depends on the preceding value of the other variable.

The VAR(3) DGP is

$$Z_{t,i} = A_1 Z_{t,i-1} + A_2 Z_{t,i-2} + A_3 Z_{t,i-3} + \Lambda \varepsilon_{t,i}, \quad (45)$$

where

$$A_1 = \begin{bmatrix} 0.55 & 0.10 \\ 0.05 & 0.50 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0.20 & 0.05 \\ 0.02 & 0.18 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0.10 & 0.02 \\ 0.01 & 0.08 \end{bmatrix}.$$

The additional univariate forecasting exercises in Section A4.4 use

$$y_{t,i} = \sum_{\ell=1}^p \phi_{\ell} y_{t,i-\ell} + \varepsilon_{t,i}.$$

The AR(1) calibration sets $\phi_1 = 0.995$, while the AR(3) calibration sets

$$(\phi_1, \phi_2, \phi_3) = (0.695, 0.20, 0.10).$$

The daily observations are aggregated as

$$\bar{Z}_t = \frac{1}{n} \sum_{i=1}^n Z_{t,i}, \quad n \in \{5, 21, 63\},$$

corresponding to weekly, monthly, and quarterly frequencies. The baseline results use period aver-

ages, while the sampling robustness exercises use the end-of-period state $Z_{t,n}$. We conduct 5,000 Monte Carlo replications. The specification exercises use samples of 20 and 40 years, while the forecast exercises use 40 years of daily observations.

For the proxy experiments, we adopt the notation of Section 2.3: $Y_{t,i}$ denotes the latent target and $X_{t,i}$ denotes any observed high-frequency companion variable. The proxy is generated as

$$Y_{t,i}^* = BY_{t,i} + \nu_{t,i}, \quad \nu_{t,i} \sim \mathcal{N}(0, r^2 \sigma_Y^2),$$

where $\nu_{t,i}$ is independent of the latent state and $r = \sigma_\nu / \sigma_Y$. The specification exercises define strong, moderate, and weak proxies using

$$(B, r) = (0.75, 0.25), \quad (0.50, 0.50), \quad (0.25, 0.75),$$

which imply proxy–target correlations of 0.949, 0.707, and 0.316, respectively. When neither the high-frequency target nor its proxy is observed, the low-frequency aggregate $\bar{Y}_t$ and the observed high-frequency predictor $X_{t,i}$ remain available.

A4.2 Specification Tests

We report the complete specification tests summarized in Section 3.1 for the triangular and feedback VAR(1) calibrations in Appendix A4.1, using weekly, monthly, and quarterly aggregation and samples of 20 and 40 years.

For each target, the baseline BUMIDAS regression conditions on the terminal high-frequency state,

$$\bar{y}_{j,t+1} = \beta_j' Z_{t,n} + u_{j,t+1},$$

where $Z_{t,n}$ contains the relevant state variables. For the autonomous y_1 equation in the triangular

DGP, $Z_{t,n}$ contains only $y_{1,t,n}$; otherwise it contains both elements of the bivariate state. State relevance is evaluated using the F-test $H_0 : \beta_j = 0$.

To test whether earlier high-frequency information remains predictive, we estimate

$$\bar{y}_{j,t+1} = \beta_j' Z_{t,n} + \gamma_j' Z_{t,n-1} + u_{j,t+1},$$

and test $H_0 : \gamma_j = 0$. We use the immediately preceding high-frequency state rather than the full set of high-frequency lags. Ghysels et al. (2020) show that conventional tests of large sets of zero restrictions can suffer from parameter proliferation and poor finite-sample performance. The one-state augmentation directly tests the VAR(1) Markov restriction without making the unrestricted regression increasingly high dimensional as the degree of aggregation rises. Conventional F-tests using larger high-frequency lag blocks give similar conclusions in the larger samples.

Low-frequency history is tested from

$$\bar{y}_{j,t+1} = \beta_j' Z_{t,n} + \delta_j' \bar{Z}_t + u_{j,t+1},$$

using the F-test $H_0 : \delta_j = 0$. For the triangular y_1 equation, only its own lagged aggregate is included; otherwise $\bar{Z}_t$ contains both aggregate state variables. Finally, a Breusch–Godfrey test of $H_0 : \text{Corr}(u_{j,t+1}, u_{j,t}) = 0$ examines whether the baseline BUMIDAS regression leaves first-order residual serial correlation that would motivate an additional low-frequency MA component.

Table A1 confirms the theoretical restrictions across DGPs, aggregation frequencies, sample lengths, and sampling schemes. The terminal BUMIDAS state is relevant in essentially every simulation, while rejection frequencies for additional high-frequency information, lagged aggregate information, and residual serial correlation remain close to the 5 percent nominal size. The same results obtain with end-of-period sampling, confirming that the specification restrictions are not

driven by period averaging.

Table A1. Specification Tests across Sampling Schemes

DGP/Target	20 Years				40 Years			
	BUMIDAS state	Extra HF state	LF AR state	BG(1)	BUMIDAS state	Extra HF state	LF AR state	BG(1)
<i>Panel A. Period Average Sampling</i>								
Weekly								
Triangular, y_1	1.000	0.047	0.049	0.053	1.000	0.049	0.046	0.049
Triangular, y_2	1.000	0.047	0.048	0.049	1.000	0.055	0.050	0.055
Feedback, y_1	1.000	0.051	0.048	0.046	1.000	0.054	0.051	0.047
Feedback, y_2	1.000	0.048	0.048	0.047	1.000	0.054	0.051	0.056
Monthly								
Triangular, y_1	1.000	0.053	0.051	0.049	1.000	0.054	0.048	0.050
Triangular, y_2	1.000	0.055	0.049	0.050	1.000	0.052	0.049	0.052
Feedback, y_1	1.000	0.054	0.047	0.047	1.000	0.049	0.048	0.049
Feedback, y_2	1.000	0.054	0.049	0.046	1.000	0.050	0.047	0.049
Quarterly								
Triangular, y_1	1.000	0.052	0.045	0.042	1.000	0.046	0.047	0.050
Triangular, y_2	1.000	0.050	0.050	0.045	1.000	0.049	0.050	0.049
Feedback, y_1	1.000	0.050	0.048	0.046	1.000	0.050	0.047	0.051
Feedback, y_2	1.000	0.053	0.048	0.047	1.000	0.050	0.046	0.051
<i>Panel B. End-of-Period Sampling</i>								
Weekly								
Triangular, y_1	1.000	0.049	0.052	0.052	1.000	0.045	0.048	0.048
Triangular, y_2	1.000	0.048	0.053	0.047	1.000	0.056	0.051	0.054
Feedback, y_1	1.000	0.049	0.048	0.050	1.000	0.052	0.052	0.049
Feedback, y_2	1.000	0.051	0.049	0.045	1.000	0.054	0.051	0.053
Monthly								
Triangular, y_1	1.000	0.050	0.055	0.055	1.000	0.054	0.048	0.048
Triangular, y_2	1.000	0.055	0.044	0.046	1.000	0.051	0.051	0.053
Feedback, y_1	1.000	0.051	0.052	0.047	1.000	0.048	0.051	0.049
Feedback, y_2	1.000	0.054	0.046	0.047	1.000	0.047	0.053	0.051
Quarterly								
Triangular, y_1	0.996	0.046	0.043	0.041	1.000	0.046	0.047	0.045
Triangular, y_2	1.000	0.050	0.043	0.044	1.000	0.048	0.048	0.050
Feedback, y_1	1.000	0.049	0.047	0.047	1.000	0.046	0.045	0.052
Feedback, y_2	1.000	0.050	0.049	0.048	1.000	0.045	0.047	0.053

Notes: Rejection frequencies from 5,000 simulations at the 5 percent level comparing period average sampling (Panel A) and end-of-period sampling (Panel B). “Extra HF state” tests $Z_{t,n-1}$; “LF AR state” tests lagged aggregate information; BG(1) tests first-order residual serial correlation.

Table A2 complements the specification tests by examining whether information criteria recover the BUMIDAS high-frequency state. The search permits aggregate AR and MA(1) terms, but the table reports the selected high-frequency histories because the aggregate terms are found to be

unnecessary when the high-frequency state is observed. The BUMIDAS specification is (1,0) for the univariate y_1 equation in the triangular design and (1,1) for the multivariate equations.

Table A2. Information Criterion Selection for Period-Averages with all High-Frequency Data Available

Aggregation	Target	Modal HF Specification [Frequency]			Mean HF Lag Counts		
		AIC	BIC	HQIC	AIC	BIC	HQIC
<i>Panel A: Triangular,</i> $(\rho_{11}, \rho_{22}, \rho_{12}, \rho_{21}, \eta_2) = (0.99, 0.99, 0, 0.50, 0.50)$							
20 Years							
Weekly	y_1	(1, 0) [55.5]	(1, 0) [98.3]	(1, 0) [88.3]	(1.51, 0.56)	(1.01, 0.01)	(1.08, 0.09)
Weekly	y_2	(1, 1) [56.5]	(1, 1) [98.5]	(1, 1) [88.7]	(1.50, 1.49)	(1.01, 1.01)	(1.08, 1.08)
Monthly	y_1	(1, 0) [53.5]	(1, 0) [95.6]	(1, 0) [82.8]	(1.56, 0.62)	(1.03, 0.02)	(1.14, 0.13)
Monthly	y_2	(1, 1) [55.4]	(1, 1) [95.1]	(1, 1) [83.0]	(1.53, 1.51)	(1.03, 1.03)	(1.14, 1.12)
Quarterly	y_1	(1, 0) [51.3]	(1, 0) [89.2]	(1, 0) [74.4]	(1.54, 0.71)	(1.04, 0.10)	(1.18, 0.28)
Quarterly	y_2	(1, 1) [51.7]	(1, 1) [89.1]	(1, 1) [75.1]	(1.56, 1.61)	(1.04, 1.09)	(1.20, 1.24)
40 Years							
Weekly	y_1	(1, 0) [55.2]	(1, 0) [98.8]	(1, 0) [90.0]	(1.51, 0.59)	(1.01, 0.01)	(1.07, 0.07)
Weekly	y_2	(1, 1) [56.6]	(1, 1) [98.7]	(1, 1) [89.5]	(1.50, 1.49)	(1.01, 1.01)	(1.06, 1.08)
Monthly	y_1	(1, 0) [55.6]	(1, 0) [97.1]	(1, 0) [86.0]	(1.51, 0.54)	(1.02, 0.02)	(1.11, 0.10)
Monthly	y_2	(1, 1) [55.0]	(1, 1) [94.9]	(1, 1) [86.1]	(1.53, 1.49)	(1.02, 1.02)	(1.11, 1.10)
Quarterly	y_1	(1, 0) [54.3]	(1, 0) [94.5]	(1, 0) [81.0]	(1.53, 0.59)	(1.03, 0.04)	(1.15, 0.17)
Quarterly	y_2	(1, 1) [54.8]	(1, 1) [93.8]	(1, 1) [81.2]	(1.55, 1.53)	(1.04, 1.04)	(1.16, 1.15)
<i>Panel B: Feedback,</i> $(\rho_{11}, \rho_{22}, \rho_{12}, \rho_{21}, \eta_2) = (0.90, 0.90, 0.10, 0.095, 0.00)$							
20 Years							
Weekly	y_1	(1, 1) [56.6]	(1, 1) [98.0]	(1, 1) [88.8]	(1.50, 1.50)	(1.01, 1.01)	(1.08, 1.08)
Weekly	y_2	(1, 1) [56.3]	(1, 1) [98.4]	(1, 1) [88.6]	(1.52, 1.49)	(1.01, 1.01)	(1.09, 1.08)
Monthly	y_1	(1, 1) [54.7]	(1, 1) [94.4]	(1, 1) [82.5]	(1.53, 1.51)	(1.03, 1.01)	(1.14, 1.13)
Monthly	y_2	(1, 1) [56.3]	(1, 1) [92.8]	(1, 1) [83.4]	(1.51, 1.50)	(1.01, 1.03)	(1.13, 1.12)
Quarterly	y_1	(1, 0) [21.3]	(1, 0) [50.3]	(1, 0) [37.3]	(1.32, 1.23)	(0.62, 0.51)	(0.83, 0.75)
Quarterly	y_2	(0, 1) [22.7]	(0, 1) [50.8]	(0, 1) [38.6]	(1.21, 1.30)	(0.51, 0.61)	(0.72, 0.84)
40 Years							
Weekly	y_1	(1, 1) [56.6]	(1, 1) [98.8]	(1, 1) [89.5]	(1.51, 1.51)	(1.01, 1.01)	(1.07, 1.08)
Weekly	y_2	(1, 1) [56.6]	(1, 1) [98.7]	(1, 1) [88.9]	(1.52, 1.50)	(1.01, 1.01)	(1.08, 1.08)
Monthly	y_1	(1, 1) [56.1]	(1, 1) [97.1]	(1, 1) [86.9]	(1.51, 1.49)	(1.02, 1.02)	(1.10, 1.09)
Monthly	y_2	(1, 1) [56.7]	(1, 1) [97.3]	(1, 1) [86.7]	(1.52, 1.49)	(1.02, 1.02)	(1.09, 1.10)
Quarterly	y_1	(1, 1) [44.3]	(1, 1) [36.1]	(1, 1) [52.5]	(1.48, 1.42)	(0.79, 0.66)	(1.02, 0.94)
Quarterly	y_2	(1, 1) [43.7]	(0, 1) [38.5]	(1, 1) [50.2]	(1.38, 1.47)	(0.63, 0.78)	(0.91, 1.02)

Notes: Modal HF specification (q_1, q_2) with selection frequency in brackets; mean selected HF lag counts are (q_1, q_2) . LF AR and MA(1) terms are allowed in the IC search. Results are based on 5,000 simulations.

The results provide clear guidance that BIC most consistently recovers the parsimonious BUMIDAS specification for both period-average and end-of-period sampling. Tables A2 and A3 show that BIC reliably recovers the BUMIDAS state in the triangular design and under weekly ag-

Table A3. Information Criterion Selection for End-of-Period Sampling with all High-Frequency Data Available

		Modal HF Specification [Frequency]			Mean HF Lag Counts					
Aggregation	Target	AIC		BIC	HQIC	AIC	BIC	HQIC		
Panel A: Triangular, $(\rho_{11}, \rho_{22}, \rho_{12}, \rho_{21}, \eta_2) = (0.99, 0.99, 0, 0.50, 0.50)$										
20 Years										
Weekly	y_1	(1, 0)	[60.1]	(1, 0)	[98.2]	(1, 0)	[88.8]	(1.30, 0.41)	(1.01, 0.01)	(1.07, 0.08)
Weekly	y_2	(1, 1)	[62.5]	(1, 1)	[98.3]	(1, 1)	[88.6]	(1.29, 1.30)	(1.01, 1.01)	(1.07, 1.07)
Monthly	y_1	(1, 0)	[59.3]	(1, 0)	[95.8]	(1, 0)	[83.3]	(1.30, 0.43)	(1.02, 0.03)	(1.10, 0.14)
Monthly	y_2	(1, 1)	[61.0]	(1, 1)	[95.2]	(1, 1)	[83.8]	(1.31, 1.31)	(1.03, 1.03)	(1.11, 1.11)
Quarterly	y_1	(1, 0)	[53.1]	(1, 0)	[87.7]	(1, 0)	[72.6]	(1.17, 0.60)	(1.00, 0.13)	(1.07, 0.31)
Quarterly	y_2	(1, 1)	[58.2]	(1, 1)	[84.6]	(1, 1)	[73.9]	(1.23, 1.37)	(0.97, 1.14)	(1.06, 1.22)
40 Years										
Weekly	y_1	(1, 0)	[59.9]	(1, 0)	[98.6]	(1, 0)	[90.3]	(1.30, 0.42)	(1.01, 0.01)	(1.06, 0.06)
Weekly	y_2	(1, 1)	[63.3]	(1, 1)	[98.8]	(1, 1)	[89.5]	(1.28, 1.29)	(1.01, 1.01)	(1.06, 1.07)
Monthly	y_1	(1, 0)	[59.9]	(1, 0)	[97.2]	(1, 0)	[86.3]	(1.30, 0.39)	(1.02, 0.01)	(1.09, 0.09)
Monthly	y_2	(1, 1)	[62.2]	(1, 1)	[97.1]	(1, 1)	[86.5]	(1.30, 1.28)	(1.02, 1.02)	(1.09, 1.08)
Quarterly	y_1	(1, 0)	[56.5]	(1, 0)	[92.0]	(1, 0)	[79.0]	(1.17, 0.52)	(1.01, 0.08)	(1.04, 0.23)
Quarterly	y_2	(1, 1)	[61.8]	(1, 1)	[93.5]	(1, 1)	[82.4]	(1.30, 1.30)	(1.01, 1.05)	(1.10, 1.13)
Panel B: Feedback, $(\rho_{11}, \rho_{22}, \rho_{12}, \rho_{21}, \eta_2) = (0.90, 0.90, 0.10, 0.095, 0.00)$										
20 Years										
Weekly	y_1	(1, 1)	[62.5]	(1, 1)	[98.2]	(1, 1)	[88.7]	(1.30, 1.30)	(1.01, 1.01)	(1.07, 1.07)
Weekly	y_2	(1, 1)	[63.2]	(1, 1)	[98.2]	(1, 1)	[89.1]	(1.29, 1.30)	(1.01, 1.01)	(1.07, 1.07)
Monthly	y_1	(1, 1)	[59.1]	(1, 1)	[64.8]	(1, 1)	[73.4]	(1.27, 1.27)	(0.86, 0.86)	(1.03, 1.04)
Monthly	y_2	(1, 1)	[56.9]	(1, 1)	[57.4]	(1, 1)	[69.0]	(1.27, 1.27)	(0.82, 0.84)	(1.01, 1.02)
Quarterly	y_1	(0, 1)	[35.0]	(0, 1)	[47.7]	(1, 0)	[43.3]	(0.78, 0.80)	(0.54, 0.54)	(0.62, 0.62)
Quarterly	y_2	(1, 0)	[34.3]	(1, 0)	[48.1]	(1, 0)	[43.3]	(0.79, 0.79)	(0.55, 0.53)	(0.63, 0.61)
40 Years										
Weekly	y_1	(1, 1)	[62.6]	(1, 1)	[98.5]	(1, 1)	[89.7]	(1.30, 1.29)	(1.01, 1.01)	(1.07, 1.06)
Weekly	y_2	(1, 1)	[63.6]	(1, 1)	[98.4]	(1, 1)	[90.0]	(1.28, 1.29)	(1.01, 1.01)	(1.06, 1.06)
Monthly	y_1	(1, 1)	[62.8]	(1, 1)	[95.4]	(1, 1)	[87.0]	(1.30, 1.29)	(1.00, 1.00)	(1.08, 1.08)
Monthly	y_2	(1, 1)	[62.7]	(1, 1)	[93.6]	(1, 1)	[86.5]	(1.29, 1.29)	(0.99, 1.00)	(1.08, 1.08)
Quarterly	y_1	(0, 1)	[33.0]	(0, 1)	[49.3]	(0, 1)	[45.3]	(0.84, 0.84)	(0.51, 0.53)	(0.58, 0.60)
Quarterly	y_2	(0, 1)	[32.9]	(0, 1)	[48.8]	(0, 1)	[44.9]	(0.82, 0.84)	(0.52, 0.53)	(0.58, 0.60)

Notes: Modal HF specification (q_1, q_2) with selection frequency in brackets; mean selected HF lag counts are (q_1, q_2) . LF AR and MA(1) terms are allowed in the IC search. Results are based on 5,000 simulations using point sampled (end-of-period) data.

gregation, while AIC generally retains more unnecessary high-frequency observations. For period averages, HQIC more often retains the weak cross-variable effects in the quarterly feedback design. With end-of-period sampling, HQIC improves recovery in the shorter monthly sample, but all three criteria frequently omit one state variable under quarterly sampling. The choice of criterion therefore depends on the sampling scheme and the strength of the predictive effects, as well as sample

size.

A4.3 Proxy Specification Tests

BUMIDAS theory shows that when the target has its own high-frequency persistence, efficient forecasting requires information about its high-frequency state. We therefore extend the simulations to quantify how the required forecast structure changes when that state is proxied or unavailable.

We extend the specification tests to the triangular VAR(1) in Appendix A4.1, using its strong, moderate, and weak target proxies and the case in which the high-frequency target is unavailable. For the univariate y_1 and multivariate y_2 targets, we test whether preceding high-frequency information adds predictive content, $H_0 : \gamma_j = 0$, and whether lagged aggregate information is redundant, $H_0 : \delta_j = 0$.

The interpretation differs from the observed-state case. Rejection of $H_0 : \gamma_j = 0$ indicates that the current-proxy approximation is insufficient, rather than that the latent VAR requires additional lags. After conditioning on the available high-frequency information and aggregate AR terms, a BG(1) test of $H_0 : \text{Corr}(u_{j,t+1}, u_{j,t}) = 0$ assesses whether residual dependence remains that would motivate a low-frequency MA component. When the high-frequency target is unavailable, only the aggregate AR and BG(1) tests apply to the univariate y_1 equation. For the multivariate y_2 equation, the high-frequency target is omitted but the observed high-frequency predictor y_1 remains available, so its current and preceding values can still enter the forecast.

For the univariate case, where neither high-frequency target data nor a proxy is available, exclusion of the aggregate AR term is consistently rejected. BG(1) also frequently rejects no residual serial correlation, although less often under quarterly aggregation, consistent with the ARMA structure induced by temporal aggregation emphasized by Weiss (1984). For the multivariate target, corresponding to a UMIDAS-type information set in which the high-frequency target is

unavailable but the high-frequency predictor remains observed, exclusion of the current predictor and aggregate AR terms is consistently rejected. Exclusion of earlier high-frequency predictor observations is also frequently rejected, and BG(1) detects residual serial correlation, consistent with Foroni et al. (2019).

Table A2 shows that BIC most consistently recovers the true BUMIDAS specification when the high-frequency state is observed. We therefore use BIC to select a finite feasible specification for the strong, moderate, and weak proxy designs. The global grid search considers zero to $p(n - 1)$ most recent high-frequency observations of each variable and zero to five aggregate AR terms. MA terms are then assessed for the BIC-minimizing specification. Table A5 reports the modal specification and mean number of selected terms across aggregation frequencies for the univariate model and the second equation of the multivariate feedback model, with and without a contemporaneous effect of the first variable on the second.

Table A4. Specification Tests with High-Frequency Proxies across Aggregation Frequencies

Noise-to-signal	Proxy strength	Corr. (y^*, y)	Univariate				Multivariate			
			Current HF info.	Extra HF state	LF AR state	BG(1)	Current HF info.	Extra HF state	LF AR state	BG(1)
<i>Panel A. Quarterly Aggregation ($n = 63$)</i>										
20 Years										
No HF target			—	—	1.000	0.340	1.000	0.212	1.000	0.089
0.25	0.75	0.95	1.000	0.639	0.277	0.151	1.000	0.999	1.000	0.082
0.50	0.50	0.71	1.000	0.958	0.991	0.960	1.000	1.000	1.000	0.949
0.75	0.25	0.32	0.662	0.580	1.000	1.000	0.933	0.871	1.000	1.000
40 Years										
No HF target			—	—	1.000	0.587	1.000	0.393	1.000	0.159
0.25	0.75	0.95	1.000	0.915	0.537	0.279	1.000	1.000	1.000	0.151
0.50	0.50	0.71	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	0.25	0.32	0.919	0.870	1.000	1.000	0.999	0.995	1.000	1.000
<i>Panel B. Monthly Aggregation ($n = 21$)</i>										
20 Years										
No HF target			—	—	1.000	0.942	1.000	0.589	1.000	0.392
0.25	0.75	0.95	1.000	1.000	1.000	0.470	1.000	1.000	1.000	0.265
0.50	0.50	0.71	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	0.25	0.32	0.998	0.993	1.000	1.000	1.000	1.000	1.000	1.000
40 Years										
No HF target			—	—	1.000	0.998	1.000	0.875	1.000	0.694
0.25	0.75	0.95	1.000	1.000	1.000	0.772	1.000	1.000	1.000	0.546
0.50	0.50	0.71	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	0.25	0.32	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
<i>Panel C. Weekly Aggregation ($n = 5$)</i>										
20 Years										
No HF target			—	—	1.000	1.000	1.000	0.997	1.000	0.993
0.25	0.75	0.95	1.000	1.000	1.000	0.940	1.000	1.000	1.000	0.980
0.50	0.50	0.71	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	0.25	0.32	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
40 Years										
No HF target			—	—	1.000	1.000	1.000	1.000	1.000	1.000
0.25	0.75	0.95	1.000	1.000	1.000	0.999	1.000	1.000	1.000	1.000
0.50	0.50	0.71	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	0.25	0.32	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Notes: Rejection frequencies from 5,000 simulations at the 5 percent level. Strong, moderate, and weak proxies correspond to $(B, \sigma_\nu/\sigma_Y) \in \{(0.75, 0.25), (0.50, 0.50), (0.25, 0.75)\}$. “Extra HF state” tests the preceding available high-frequency state; “LF AR state” tests lagged aggregate information; BG(1) tests first-order residual serial correlation. Aggregation levels correspond to $n = 63$ (quarterly), $n = 21$ (monthly), and $n = 5$ (weekly).

Table A5. BIC-Selected Specifications with High-Frequency Proxies

Modal Specification			Mean Number Selected			
Frequency	Proxy (Corr.)	[Frequency]	HF predictor	HF proxy	LF AR	LF MA
Panel A. Univariate						
Quarterly	Strong (0.95)	(2, 0, 0) [49.1]	—	2.266	0.125	0.004
	Moderate (0.71)	(3, 1, 0) [17.2]	—	3.526	0.954	0.084
	Weak (0.32)	(0, 1, 0) [33.9]	—	0.516	1.489	0.052
Monthly	Strong (0.95)	(3, 1, 0) [27.4]	—	3.313	0.774	0.089
	Moderate (0.71)	(3, 2, 0) [24.1]	—	2.796	1.989	0.192
	Weak (0.32)	(0, 2, 0) [68.3]	—	0.265	2.114	0.036
Weekly	Strong (0.95)	(4, 2, 1) [33.6]	—	3.288	2.269	0.520
	Moderate (0.71)	(1, 2, 0) [36.4]	—	0.811	2.403	0.127
	Weak (0.32)	(0, 2, 0) [49.8]	—	0.058	2.419	0.069
Panel B. Multivariate, Lagged Dynamic Effects Only						
Quarterly	Strong (0.95)	(1, 1, 0, 0) [41.3]	1.062	0.978	0.156	0.000
	Moderate (0.71)	(1, 0, 0, 0) [58.7]	1.102	0.176	0.267	0.008
	Weak (0.32)	(1, 0, 0, 0) [64.3]	1.114	0.054	0.287	0.001
Monthly	Strong (0.95)	(1, 2, 1, 0) [29.7]	1.060	2.397	0.924	0.074
	Moderate (0.71)	(1, 0, 2, 0) [27.7]	1.052	0.543	1.469	0.186
	Weak (0.32)	(1, 0, 2, 0) [37.7]	1.048	0.048	1.493	0.208
Weekly	Strong (0.95)	(1, 1, 2, 0) [55.7]	1.070	1.804	2.114	0.028
	Moderate (0.71)	(1, 0, 2, 0) [50.1]	1.080	0.365	2.154	0.008
	Weak (0.32)	(1, 0, 2, 0) [76.4]	1.078	0.026	2.148	0.004
Panel C. Multivariate, with Contemporaneous Impact						
Quarterly	Strong (0.95)	(1, 0, 0, 0) [76.2]	1.034	0.212	0.048	0.000
	Moderate (0.71)	(1, 0, 0, 0) [88.4]	1.036	0.070	0.048	0.000
	Weak (0.32)	(1, 0, 0, 0) [90.8]	1.034	0.038	0.048	0.000
Monthly	Strong (0.95)	(1, 2, 0, 0) [23.4]	1.028	1.597	0.433	0.036
	Moderate (0.71)	(1, 0, 1, 0) [41.3]	1.022	0.180	0.842	0.096
	Weak (0.32)	(1, 0, 1, 0) [46.9]	1.022	0.028	0.862	0.096
Weekly	Strong (0.95)	(3, 1, 2, 0) [34.5]	3.056	1.196	2.126	0.156
	Moderate (0.71)	(3, 0, 2, 0) [39.9]	3.180	0.136	2.154	0.078
	Weak (0.32)	(3, 0, 2, 0) [45.3]	3.182	0.014	2.150	0.074

Notes: Entries under “Modal Specification” report the most frequently selected lag structure, with its percentage selection frequency in brackets. A univariate specification (p, p_{ar}, q_{ma}) selects p high-frequency proxy observations alongside p_{ar} aggregate AR and q_{ma} aggregate MA terms. A multivariate specification $(q_1, q_2, p_{AR}, q_{MA})$ reports the numbers of observed high-frequency predictor observations, target-proxy observations, aggregate AR terms, and aggregate MA terms, respectively. Proxy strength correlation is $\text{Corr}(y^*, y)$. 40 years of data, temporal aggregation $n \in \{5, 21, 63\}$, and simulations based on the corresponding DGPs.

A4.4 Univariate Forecasts

For the forecasts, each Monte Carlo replication uses a 40-year sample, approximating the span of daily data available in many macro-financial applications. The first 75 percent of the sample is used for estimation and the remaining 25 percent for out-of-sample evaluation. When the UMIDAS or Extra HF lag window has more coefficients than low-frequency estimation observations, as under quarterly aggregation with a VAR(3), AR(3), or proxy, it is truncated to the most recent estimable high-frequency lags.

MSFE ratios use the period-average no-change forecast as the benchmark. Success ratios measure the proportion of correctly predicted changes in the period-average target. Where reported, bracketed entries give Monte Carlo rejection frequencies from one-sided Diebold–Mariano tests (Diebold and Mariano, 1995), using squared-error loss differentials and Newey–West HAC standard errors (Newey and West, 1987).

First consider daily univariate DGPs of an AR(1) with coefficient 0.995, or an AR(3), with coefficients (0.695, 0.20, 0.10). BUMIDAS conditions on the DGP-implied daily AR state, while BUMIDAS-BIC selects its high-frequency lag order in real-time using BIC. Bottom-Up Recursive estimates the model at the daily frequency, recursively forecasts the daily path, and averages those forecasts to the target frequency (Zellner and Montmarquette, 1971; Lütkepohl, 1986; Benmoussa et al., 2026). The end-of-period (EoP) no-change forecast is included as a naive benchmark since it is the optimal forecast under the random walk null (Ellwanger and Snudden, 2023; McCarthy and Snudden, 2025). Extra HF Terms includes $p(n - 1)$ unrestricted high-frequency observations, while Almon imposes quadratic lag weights estimated over $(n - 1)$ high-frequency observations. LF ARMA uses only aggregate observations. Because the high-frequency state is observed, the mixed-frequency models exclude additional aggregate AR and MA terms; these are introduced when the target is proxied.

Table A6. Univariate Forecast Performance and DM Tests with Observed High-Frequency Data

DGP	Freq.	BUMIDAS	BUMIDAS BIC	Bottom-Up Recursive	Extra HF Terms	Almon	LF ARMA	EoP NC
Data		Mixed	Mixed	Daily	Mixed	Mixed	Aggregate	Daily
<i>Panel A. Relative MSFE</i>								
AR(1)	Quarterly	0.529	0.530	0.527	1.140 [97]	0.618 [54]	0.916 [93]	0.589 [54]
AR(1)	Monthly	0.540	0.540	0.540	0.573 [53]	0.605 [74]	0.969 [100]	0.561 [53]
AR(1)	Weekly	0.647	0.647	0.647	0.649 [25]	0.653 [40]	0.992 [100]	0.654 [50]
AR(3)	Quarterly	0.547	0.543	0.535	8.86 [100]	0.622 [45]	0.911 [90]	0.585 [36]
AR(3)	Monthly	0.569	0.572	0.540	0.682 [86]	0.613 [58]	0.928 [100]	0.595 [46]
AR(3)	Weekly	0.746	0.748	0.745	0.752 [36]	0.747 [20]	0.956 [100]	0.801 [95]
<i>Panel B. Success Ratio</i>								
AR(1)	Quarterly	0.745	0.744	0.746	0.657	0.719	0.598	0.729
AR(1)	Monthly	0.740	0.740	0.740	0.729	0.719	0.559	0.733
AR(1)	Weekly	0.703	0.703	0.703	0.702	0.701	0.529	0.701
AR(3)	Quarterly	0.740	0.741	0.743	0.558	0.717	0.605	0.730
AR(3)	Monthly	0.730	0.729	0.740	0.702	0.716	0.591	0.724
AR(3)	Weekly	0.668	0.668	0.669	0.667	0.668	0.568	0.662

Notes: Mean MSFE ratios and success ratios (5,000 replications, $T = 40$ years). Bracketed values are percentage rejections of a one-sided DM test (10% level) favoring BUMIDAS. Quarterly Extra HF lags are truncated to those estimable in the initial sample. The daily AR(1) or AR(3) target is aggregated to $n \in \{5, 21, 63\}$.

Table A6 shows that large forecast gains arise from using high-frequency information efficiently. BUMIDAS and the recursive bottom-up forecast deliver nearly identical MSFEs, while BUMIDAS-BIC performs similarly when the high-frequency lag order is selected rather than known. Relative to the aggregate no-change and low-frequency AR benchmarks, these methods reduce MSFE by roughly 39–48 percent across monthly and quarterly aggregation. They also outperform the end-of-period no-change forecast, improving upon the random-walk null and demonstrating the usefulness of model-based forecasts.

The cost of estimating unnecessary high-frequency history increases sharply with the degree of aggregation. Extra HF Terms and Almon remain close to the parsimonious forecasts at the weekly frequency but become less accurate under monthly and especially quarterly aggregation. This arises both from parameter proliferation as well as the shorter number of observations in the 40-year quarterly sample versus weekly sampling. At the quarterly frequency, the unrestricted

specification performs worse than even the aggregate no-change forecast under both DGPs and becomes particularly unstable under the AR(3). Almon restrictions substantially mitigate this parameter-proliferation cost but do not eliminate it.

The bracketed DM rejection frequencies confirm that the economically important differences are generally detectable. At the quarterly frequency, equal predictive accuracy is rejected in favor of BUMIDAS in 90–93 percent of replications against LF ARMA and 97–100 percent against Extra HF Terms. Rejection frequencies are lower against the Almon specification, consistent with the lag restrictions reducing the efficiency loss from estimating unnecessary high-frequency history.

Table A7 isolates how persistence and temporal aggregation affect the value of high-frequency information. We consider $\rho \in 0.90, 0.95, 0.98, 0.99$, a range typical for daily macro-financial series. Unlike Tiao (1972), who stopped at $\rho \leq 0.90$, we therefore provide insights into the high-persistence boundary. A key insight is that the upper range of these persistences produces large losses in forecast accuracy from aggregation akin to a unit-root process.

Across persistence, BUMIDAS produces consistently large MSFE reductions relative to the aggregate no-change forecast, while its gains relative to LF ARMA increase with persistence. Moving from $\rho = 0.90$ to $\rho = 0.99$ raises the gain of BUMIDAS relative to LF ARMA from approximately 25 to 33 percent weekly, from 20 to 41 percent monthly, and from 8 to 38 percent quarterly. Across all persistence values, BUMIDAS is essentially tied with Extra HF Terms and Almon weekly but increasingly outperforms both as aggregation lengthens, with Extra HF Terms incurring the largest quarterly loss. Persistence therefore magnifies the value of high-frequency information relative to aggregate forecasts, while aggregation magnifies the cost of unnecessary high-frequency history.

Table A8 extends the univariate exercise to the case in which the daily target is latent and only its low-frequency aggregate and a high-frequency proxy are observed, consistent with Section 2.3. The proxy calibrations are those defined in Appendix A4.1. For comparison, the Last- p Proxy

Table A7. AR(1), One-Period-Ahead Forecast Performance across Persistence and Aggregation

ρ	BUMIDAS	Bottom-Up Recursive	Extra HF Terms	Almon	LF ARMA	EoP No-change
<i>Panel A. Quarterly</i>						
0.990	0.528	0.525	1.139	0.604	0.847	0.657
0.980	0.527	0.524	1.139	0.594	0.750	0.805
0.950	0.524	0.521	1.140	0.572	0.624	1.299
0.900	0.523	0.518	1.138	0.554	0.567	2.210
<i>Panel B. Monthly</i>						
0.990	0.539	0.539	0.572	0.596	0.919	0.582
0.980	0.537	0.537	0.570	0.591	0.863	0.626
0.950	0.532	0.531	0.564	0.577	0.756	0.768
0.900	0.526	0.525	0.558	0.559	0.654	1.035
<i>Panel C. Weekly</i>						
0.990	0.646	0.646	0.648	0.645	0.960	0.659
0.980	0.643	0.643	0.645	0.642	0.932	0.670
0.950	0.635	0.635	0.637	0.634	0.888	0.703
0.900	0.623	0.623	0.625	0.622	0.832	0.762

Notes: Mean MSFE ratios relative to the aggregate no-change forecast (5,000 replications, $T = 40$ years). The latent daily AR(1) target is aggregated to weekly ($n = 5$), monthly ($n = 21$), or quarterly ($n = 63$) frequencies. Extra HF Terms include $p(n - 1)$ unrestricted high-frequency terms. LF and EoP denote low-frequency and end-of-period.

Projection uses only the p most recent proxy observations. BUMIDAS-BIC globally selects the high-frequency proxy history and aggregate AR order using BIC. Extra HF Terms and Almon retain their respective unrestricted and restricted high-frequency parameterizations and use aggregate AR orders commensurate with the daily lags, while LF AR selects among aggregate lags using BIC.

Table A8 shows that efficient use of high-frequency information reduces MSFE relative to low-frequency data alone only when the proxy is sufficiently informative. With a strong proxy, BUMIDAS-BIC outperforms LF AR in every design, with the DM test favoring BUMIDAS-BIC in 76–94 percent of replications. The Last- p Proxy Projection remains competitive under quarterly aggregation but deteriorates at the monthly and weekly frequencies even with a strong proxy, with the losses increasing sharply as the proxy weakens. BUMIDAS-BIC substantially limits these losses by selecting additional proxy and aggregate lags, consistent with earlier proxy observations

Table A8. Univariate Forecast Performance and DM Tests with High-Frequency Proxies

Frequency	Proxy (Corr.)	Last- p	BUMIDAS BIC	Extra HF Terms	Almon	LF AR
<i>Panel A. AR(1)</i>						
Quarterly	Strong (0.95)	0.663	0.612	1.258 [94]	0.634 [20]	0.912 [78]
	Moderate (0.71)	1.676	0.814	1.529 [87]	0.718 [3]	0.912 [39]
	Weak (0.32)	2.566	0.924	1.871 [91]	0.866 [6]	0.912 [12]
Monthly	Strong (0.95)	0.939	0.689	0.697 [17]	0.675 [6]	0.922 [94]
	Moderate (0.71)	4.034	0.898	0.928 [27]	0.884 [7]	0.922 [27]
	Weak (0.32)	6.822	0.948	1.015 [52]	0.963 [20]	0.922 [4]
Weekly	Strong (0.95)	2.312	0.879	0.904 [70]	0.919 [85]	0.938 [92]
	Moderate (0.71)	15.243	0.936	0.987 [90]	0.988 [90]	0.938 [18]
	Weak (0.32)	26.960	0.941	0.994 [91]	0.993 [90]	0.938 [5]
<i>Panel B. AR(3)</i>						
Quarterly	Strong (0.95)	0.621	0.633	9.252 [100]	0.642 [16]	0.930 [76]
	Moderate (0.71)	1.387	0.845	11.858 [100]	0.748 [3]	0.930 [35]
	Weak (0.32)	3.055	0.948	17.697 [100]	0.901 [7]	0.930 [10]
Monthly	Strong (0.95)	0.781	0.720	0.830 [67]	0.699 [4]	0.930 [91]
	Moderate (0.71)	3.030	0.919	1.056 [70]	0.911 [9]	0.930 [19]
	Weak (0.32)	8.045	0.961	1.115 [79]	0.976 [20]	0.930 [3]
Weekly	Strong (0.95)	1.582	0.914	0.916 [21]	0.944 [72]	0.957 [80]
	Moderate (0.71)	10.423	0.961	0.960 [7]	0.992 [67]	0.957 [5]
	Weak (0.32)	30.228	0.965	0.964 [6]	0.996 [66]	0.957 [2]

Notes: Mean MSFE ratios relative to the period-average no-change forecast (5,000 replications, $T = 40$ years). Bracketed values are percentage rejection frequencies of a one-sided Diebold–Mariano test (10% level) in favor of BUMIDAS-BIC. The latent daily target follows an AR(1) or AR(3), aggregated to $n \in \{5, 21, 63\}$.

becoming informative when the current proxy does not adequately recover the latent state.

The cost of Extra HF Terms rises sharply with aggregation, particularly quarterly, where the DM test favors BUMIDAS-BIC in 87–100 percent of replications. Almon avoids much of this deterioration and yields lower MSFE than BUMIDAS-BIC with moderate and weak quarterly proxies, indicating gains from pooling a longer proxy history. At the monthly frequency, Almon and BUMIDAS-BIC perform similarly with strong and moderate proxies, whereas LF AR performs best with a weak proxy. Extra HF Terms is more competitive at the weekly frequency, when relatively few unrestricted coefficients are required, although BUMIDAS-BIC generally remains the best or nearly best high-frequency specification. With a weak weekly proxy, LF AR performs

best and Almon performs worse than LF AR, suggesting that a sufficiently noisy proxy provides little incremental information and that longer lag distributions can accumulate noise. Overall, BUMIDAS-BIC provides the most robust high-frequency performance across proxy qualities and aggregation frequencies, while Almon can be preferred when pooling a longer proxy history is particularly valuable.

A4.5 Multivariate Forecasts

This subsection complements the results in Section 3.2.

Mean directional accuracy, or success ratios, for the same multivariate simulations are provided: Table A9 considers the case in which the high-frequency target is observed, while Table A10 considers the cases in which the target is observed through a noisy daily proxy or is unavailable.

Table A9. Directional Accuracy when High-Frequency Data of the Target is Observed

Method	Weekly	Monthly	Quarterly
<i>Panel A. VAR(1)</i>			
BU-Recursive	0.700	0.732	0.742
BUMIDAS	0.699	0.732	0.741
BUMIDAS-BIC	0.699	0.732	0.741
UMIDAS	0.633	0.696	0.644
RMIDAS	0.632	0.690	0.706
LF VAR	0.559	0.546	0.568
EoP NC	0.683	0.717	0.728
<i>Panel B. VAR(3)</i>			
BU-Recursive	0.685	0.727	0.743
BUMIDAS	0.685	0.725	0.735
BUMIDAS-BIC	0.684	0.725	0.740
UMIDAS	0.644	0.670	0.555
RMIDAS	0.646	0.698	0.725
LF VAR	0.644	0.680	0.713
EoP NC	0.609	0.626	0.606

Notes: Mean directional accuracy, defined as the proportion of forecasts that correctly predict the direction of change in the target (5,000 replications, $T = 40$ years). The daily target follows a VAR(1) or VAR(3), aggregated to $n \in \{5, 21, 63\}$. BU-Recursive averages daily-VAR forecasts; LF and EoP NC denote the low-frequency and end-of-period no-change forecasts.

Tables A9 and A10 show that the broad MSFE rankings extend to directional accuracy. When

the high-frequency target is observed, BU-Recursive, BUMIDAS, and BUMIDAS-BIC have nearly identical success ratios and achieve the highest directional accuracy across both DGPs. When the target is proxied or unavailable, BUMIDAS-BIC matches or closely approaches the highest success ratio at monthly and quarterly frequencies and consistently exceeds UMIDAS, RMIDAS, and LF VAR. RMIDAS is generally the closest conventional alternative, while differences among the mixed-frequency methods are small at the weekly frequency. Proxy quality matters most at the monthly and weekly frequencies, with the weak-proxy results often similar to those obtained when the target is unavailable.

Table A11 assesses the relative forecasting performance of alternative information criteria (BIC, AIC, HQIC) in the presence of a high-frequency proxy. Generally, the differences in performance are marginal. These results motivate the use of BIC as the baseline.

Table A12 repeats the observed high-frequency target exercise of Section 3.2 under end-of-period sampling. The rankings of the high- and mixed-frequency methods are unchanged. BUMIDAS and BUMIDAS-BIC closely match the bottom-up forecast at every frequency, while UMIDAS deteriorates sharply with aggregation and VAR order, and equal predictive accuracy is rejected in favor of BUMIDAS-BIC in 96–100 percent of replications at the quarterly frequency and 52–85 percent at the monthly frequency. End-of-period sampling mainly changes the performance of the low-frequency VAR. Because the low-frequency observation is now the terminal high-frequency state $Z_{t,n}$, under the VAR(1) the low-frequency VAR projects on the BUMIDAS state and delivers identical forecasts, with no DM rejections. Under the VAR(3), it omits $Z_{t,n-1}$ and $Z_{t,n-2}$, so small losses remain, and DM tests reject in 13–35 percent of replications. RMIDAS stays close to BUMIDAS-BIC, with rejection frequencies of 14–34 percent. Success ratios yield the same ranking.

Table A10. Directional Accuracy when the High-Frequency Target is Proxied or Unavailable

Frequency	Proxy (Corr.)	Last- p	BUMIDAS BIC	UMIDAS	RMIDAS	LF VAR
<i>Panel A. VAR(1)</i>						
Quarterly	Strong (0.95)	0.733	0.733	0.533	0.704	0.566
	Moderate (0.71)	0.731	0.730	0.534	0.701	0.566
	Weak (0.32)	0.731	0.730	0.533	0.701	0.566
	Unavailable	0.734	0.734	0.644	0.706	0.568
Monthly	Strong (0.95)	0.712	0.712	0.697	0.700	0.546
	Moderate (0.71)	0.705	0.704	0.687	0.690	0.546
	Weak (0.32)	0.704	0.704	0.686	0.689	0.546
	Unavailable	0.704	0.704	0.696	0.690	0.546
Weekly	Strong (0.95)	0.631	0.644	0.645	0.642	0.559
	Moderate (0.71)	0.618	0.637	0.634	0.632	0.559
	Weak (0.32)	0.618	0.636	0.633	0.632	0.559
	Unavailable	0.617	0.632	0.633	0.632	0.559
<i>Panel B. VAR(3)</i>						
Quarterly	Strong (0.95)	0.733	0.737	0.567	0.722	0.713
	Moderate (0.71)	0.731	0.734	0.567	0.722	0.713
	Weak (0.32)	0.729	0.732	0.562	0.720	0.713
	Unavailable	0.732	0.735	0.555	0.725	0.713
Monthly	Strong (0.95)	0.723	0.722	0.665	0.715	0.680
	Moderate (0.71)	0.713	0.711	0.655	0.709	0.680
	Weak (0.32)	0.702	0.702	0.644	0.700	0.680
	Unavailable	0.701	0.701	0.670	0.698	0.680
Weekly	Strong (0.95)	0.677	0.677	0.676	0.678	0.644
	Moderate (0.71)	0.648	0.654	0.655	0.656	0.644
	Weak (0.32)	0.641	0.646	0.643	0.647	0.644
	Unavailable	0.643	0.643	0.644	0.646	0.644

Notes: Mean directional accuracy, defined as the proportion of forecasts that correctly predict the direction of change in the target (5,000 replications, $T = 40$ years). The daily target follows a VAR(1) or VAR(3), aggregated to $n \in \{5, 21, 63\}$. Corr. denotes the correlation between the high-frequency proxy and the latent daily target. Bold indicates the highest directional accuracy among BUMIDAS-BIC, UMIDAS, RMIDAS, and the LF VAR; Last- p is excluded as a known-order benchmark.

Table A11. Forecast Performance and Directional Accuracy Using Alternative Information Criteria

Frequency	Proxy (Corr.)	VAR(1)			VAR(3)		
		BIC	HQIC	AIC	BIC	HQIC	AIC
Panel A. Mean Squared Forecast Error (MSFE)							
Quarterly	Strong (0.95)	0.568	0.570	0.574	0.554	0.557	0.560
	Moderate (0.71)	0.576	0.578	0.582	0.565	0.567	0.571
	Weak (0.32)	0.576	0.579	0.583	0.572	0.575	0.579
Monthly	Strong (0.95)	0.629	0.627	0.625	0.591	0.590	0.590
	Moderate (0.71)	0.656	0.654	0.654	0.630	0.627	0.625
	Weak (0.32)	0.658	0.656	0.655	0.656	0.654	0.654
Panel B. Success Ratio							
Quarterly	Strong (0.95)	0.733	0.732	0.732	0.737	0.736	0.736
	Moderate (0.71)	0.730	0.729	0.728	0.734	0.733	0.732
	Weak (0.32)	0.730	0.729	0.728	0.732	0.731	0.730
Monthly	Strong (0.95)	0.712	0.713	0.713	0.722	0.723	0.722
	Moderate (0.71)	0.704	0.704	0.703	0.711	0.711	0.711
	Weak (0.32)	0.704	0.703	0.703	0.702	0.702	0.702

Notes: Mean MSFE ratios and success ratios of forecasts utilizing alternative information criteria to infer the high-frequency state specification. Results are based on 5,000 replications over a 40-year sample. The latent daily target follows a VAR(1) or VAR(3), aggregated to $n \in \{21, 63\}$. Corr. denotes the correlation between the high-frequency proxy and the latent daily target.

Table A12. Multivariate Forecast Performance and DM Tests End-of-Period Sampling

DGP	Freq.	Bottom-Up	BUMIDAS	BUMIDAS BIC	UMIDAS	RMIDAS	LF VAR
<i>Panel A. Relative MSFE</i>							
VAR(1)	Quarterly	0.922	0.930	0.933	2.056 [96]	0.952 [23]	0.930 [0]
VAR(1)	Monthly	0.934	0.937	0.938	0.993 [52]	0.951 [29]	0.937 [0]
VAR(1)	Weekly	0.928	0.928	0.928	0.931 [24]	0.930 [20]	0.928 [0]
VAR(3)	Quarterly	0.526	0.550	0.532	8.716 [100]	0.550 [14]	0.550 [13]
VAR(3)	Monthly	0.583	0.592	0.587	0.713 [85]	0.594 [14]	0.593 [14]
VAR(3)	Weekly	0.697	0.699	0.701	0.709 [49]	0.704 [34]	0.705 [35]
<i>Panel B. Success Ratio</i>							
VAR(1)	Quarterly	0.595	0.593	0.592	0.529	0.583	0.593
VAR(1)	Monthly	0.584	0.584	0.583	0.564	0.574	0.584
VAR(1)	Weekly	0.587	0.587	0.587	0.586	0.586	0.587
VAR(3)	Quarterly	0.746	0.739	0.745	0.559	0.740	0.740
VAR(3)	Monthly	0.725	0.722	0.724	0.692	0.722	0.722
VAR(3)	Weekly	0.686	0.685	0.685	0.682	0.683	0.683

Notes: Mean MSFE ratios and success ratios relative to the end-of-period no-change forecast (5,000 replications, $T = 40$ years). Bracketed values are percentage rejections of a one-sided DM test favoring BUMIDAS-BIC. The daily VAR(1) or VAR(3) target is end-of-period sampled with $n \in \{5, 21, 63\}$.

A5 Forecast Application Details and Robustness

A5.1 Forecast Evaluation

We report two common forecast evaluation criteria: the MSFE ratio and the success ratio for directional accuracy. Forecasts target the period average and are expressed relative to the period-average no-change forecast. Robustness that expresses relative to the end-of-period no-change (as per, Ellwanger and Snudden, 2023; McCarthy and Snudden, 2025) replaces $\bar{y}_m$ with $y_{m,n}$.

Let $\bar{y}_{m+h}$ denote the realized period average for target y over horizon h , and let $\hat{y}_{m+h|m}$ denote the forecast made at time m for $\bar{y}_{m+h}$.

$$MSFE_h^{ratio} = \frac{\frac{1}{T} \sum_{m=1}^T (\bar{y}_{m+h} - \hat{y}_{m+h|m})^2}{\frac{1}{T} \sum_{m=1}^T (\bar{y}_{m+h} - \bar{y}_m)^2}. \quad (46)$$

Where appropriate, we also conduct Diebold-Mariano (DM) tests (Diebold and Mariano, 1995) of equal forecast accuracy. For the DM tests, loss differentials are defined as differences in squared forecast errors, and their long-run variances are estimated using the Newey–West HAC estimator (Newey and West, 1987) with truncation lag $L = \text{round } 4(P/100)^{2/9}$ where P denotes the number of out-of-sample observations. This bandwidth is a rounded implementation of the rule of thumb in Newey and West (1994).

Mean directional accuracy is denoted as the success ratio, SR_h , and measures the proportion of times the model correctly predicts the direction of change in the target variable:

$$SR_h = \frac{1}{T} \sum_{m=1}^T \mathbf{1} [\text{sgn}(\bar{y}_{m+h} - \bar{y}_m) = \text{sgn}(\hat{y}_{m+h|m} - \bar{y}_m)], \quad (47)$$

where $\mathbf{1}[\cdot]$ is the indicator function. We test for directional predictability using the independence test developed by Pesaran and Timmermann (2009).

A5.2 3-Month Treasury Bill

Table A13. Directional Accuracy for the Nominal 3-Month Treasury-Bill Rate

Method	Direct	Bottom-Up	BUMIDAS	Extra HF	Almon	EoP
Data	Average	Recursive Daily	Mixed	Terms Mixed	Mixed	No-change Daily
<i>Panel A. Monthly Forecasts</i>						
Horizon (months)	Success Ratios					
1	0.47 (–)	0.76 (< 0.001)	0.73 (< 0.001)	0.67 (< 0.001)	0.70 (< 0.001)	0.75 (< 0.001)
3	0.51 (–)	0.69 (< 0.001)	0.61 (< 0.001)	0.61 (< 0.001)	0.61 (< 0.001)	0.68 (< 0.001)
12	0.54 (–)	0.65 (< 0.001)	0.59 (< 0.001)	0.64 (< 0.001)	0.59 (0.006)	0.64 (< 0.001)
<i>Panel B. Quarterly Forecasts</i>						
Horizon (quarters)	Success Ratios					
1	0.55 (–)	0.77 (< 0.001)	0.78 (< 0.001)	0.54 (0.279)	0.65 (< 0.001)	0.77 (< 0.001)
2	0.54 (–)	0.73 (< 0.001)	0.68 (< 0.001)	0.53 (0.368)	0.66 (< 0.001)	0.72 (< 0.001)
4	0.54 (–)	0.72 (< 0.001)	0.63 (0.001)	0.59 (0.076)	0.65 (0.001)	0.71 (< 0.001)

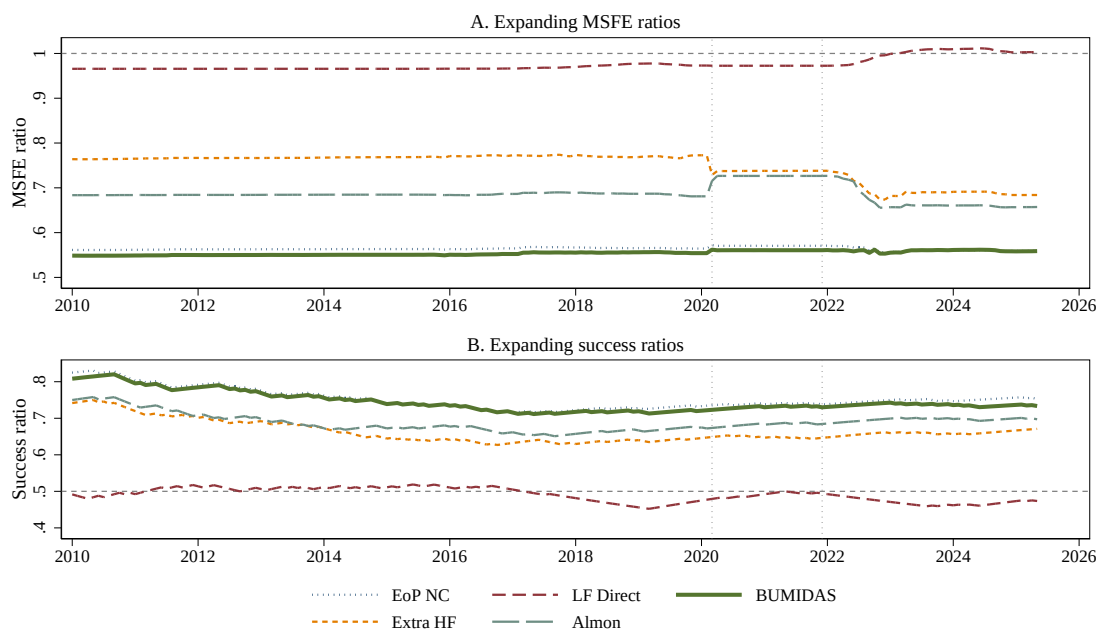
Notes: Corresponding success ratios for the forecasts reported in Table 6, 2000M1–2025M6 and 2000Q1–2025Q2. Pesaran and Timmermann (2009) p -values in parentheses.

Table A14. Real-Time Model-based Forecasts of the Nominal 3-Month Treasury Bill Rate vs End-of-Period No-Change

Method	Direct	Bottom-up	BUMIDAS	Extra HF	Almon
Data	Average	Daily	Mixed	Mixed	Mixed
horizon	Monthly Forecasts				
1	1.79 (1.00)	1.00 (0.63)	1.00 (0.47)	1.22 (0.91)	1.17 (0.93)
3	1.20 (0.99)	1.00 (0.74)	0.99 (0.37)	1.06 (0.81)	0.96 (0.26)
12	0.96 (0.25)	1.00 (0.47)	0.91 (0.08)	0.93 (0.22)	0.88 (0.02)
horizon	Quarterly Forecasts				
1	2.50 (0.99)	1.01 (0.95)	1.03 (0.63)	4.64 (0.98)	1.28 (0.91)
2	1.40 (0.98)	1.00 (0.95)	0.98 (0.38)	2.84 (0.99)	0.98 (0.41)
4	1.08 (0.74)	1.00 (0.96)	0.92 (0.20)	2.20 (0.99)	0.83 (0.08)

Note: Real-time, out-of-sample forecasts of the nominal monthly and quarterly average 3-month Treasury Bill rate, ending in 2025M6. MSFE ratios are relative to the corresponding end-of-period no-change forecast, with one-sided Diebold–Mariano p -values reported in parentheses.

Figure A1. Evolution of One-Step-Ahead Treasury Bill Forecast Performance



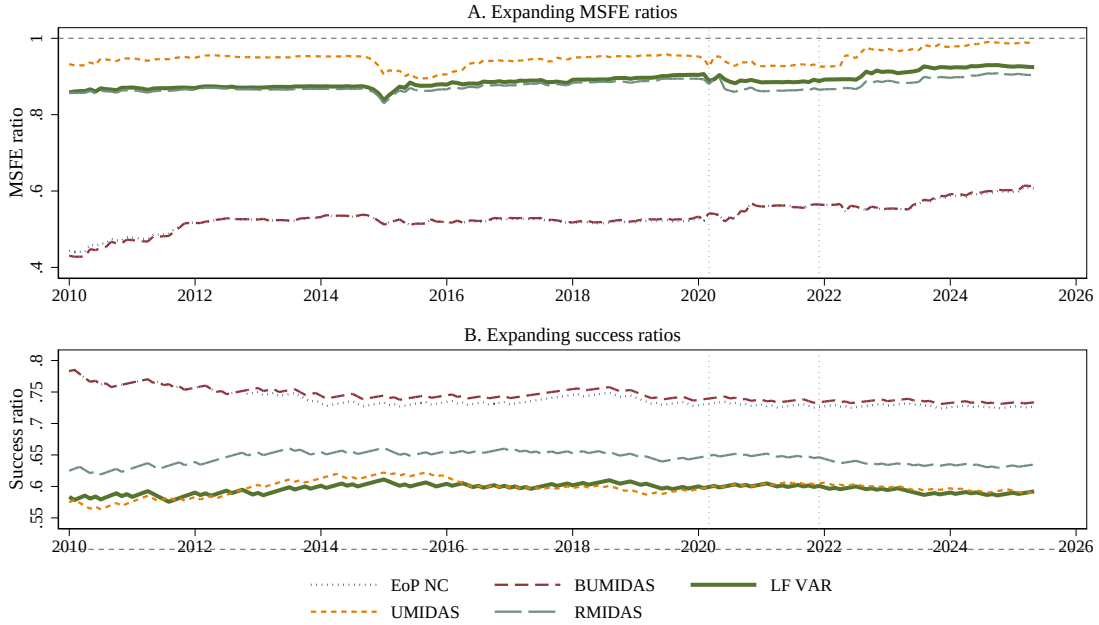
Notes: Expanding one-step-ahead MSFE ratios (Panel A) and success ratios (Panel B). MSFEs are relative to the period-average no-change forecast. Paths begin in 2010 to reduce small-sample variability; vertical lines delimit the COVID-19 period (2020M3–2021M12).

A5.3 Exchange rate forecasts

We assess the robustness of the exchange-rate forecasts in Section 4.2 to alternative transformations, lag orders, exchange-rate measures, and temporal sampling.

Figure A2 shows that the relative gains are robust across the forecast evaluation sample.

Figure A2. Evolution of One-Month-Ahead NOK/USD Forecast Performance



Notes: Expanding one-step-ahead NOK/USD MSFE ratios (Panel A) and Success Ratios (Panel B) are relative to the period-average no-change forecast. Paths begin in 2010 to reduce small-sample variability; vertical lines delimit the COVID-19 period (2020M3–2021M12).

Table A15 shows that conventional specification changes have little effect. Panel A estimates the models in levels using log real oil prices and the level of nominal exchange rates, Panel B changes transformations of the daily predictors to month-over-month growth rates, and Panel C uses VAR(3) dynamics (consistent with HQIC). The CAD and NOK results are also reproduced using the BIS narrow nominal effective exchange rates; see data Appendix A1.

Across these alternatives, the MSFE ratios for BUMIDAS change little. Longer lag structures generally worsen forecast accuracy, with a larger deterioration for unrestricted UMIDAS than for RMIDAS. The same patterns hold for the BIS narrow nominal effective exchange rates. These

specification choices have relatively modest effects compared with the gains from making high-frequency exchange-rate information available to the forecast models and imposing a parsimonious structure on that information.

Table A15. Robustness of Monthly-Average Exchange Rate Forecasts

Method	CAD	NOK	CADNEER	NOKNEER
<i>Panel A. In levels (log real oil prices)</i>				
EoP NC	0.66 [0.000]	0.61 [0.000]	0.72 [0.000]	0.64 [0.000]
BUMIDAS	0.66 [0.000]	0.60 [0.000]	0.72 [0.000]	0.67 [0.000]
LF VAR	1.02 [0.783]	1.03 [0.945]	1.02 [0.753]	1.05 [0.972]
UMIDAS	0.99 [0.416]	0.98 [0.353]	1.02 [0.763]	1.05 [0.961]
RMIDAS	0.98 [0.139]	0.98 [0.154]	0.99 [0.180]	1.01 [0.651]
BUMIDAS-X	0.66 [0.000]	0.61 [0.000]	0.73 [0.000]	0.64 [0.000]
UMIDAS-HF	0.79 [0.000]	0.70 [0.000]	0.83 [0.021]	0.88 [0.082]
RMIDAS-HF	0.75 [0.000]	0.70 [0.000]	0.79 [0.000]	0.75 [0.000]
<i>Panel B. Month-over-month growth rates for daily lags</i>				
BUMIDAS	0.66 [0.000]	0.61 [0.000]	0.73 [0.000]	0.64 [0.000]
LF VAR	0.93 [0.000]	0.92 [0.051]	0.95 [0.034]	0.95 [0.071]
UMIDAS	1.02 [0.641]	1.05 [0.684]	1.01 [0.620]	1.03 [0.603]
RMIDAS	0.94 [0.012]	0.92 [0.053]	0.96 [0.102]	0.94 [0.021]
BUMIDAS-X	0.66 [0.000]	0.61 [0.000]	0.72 [0.000]	0.63 [0.000]
UMIDAS-HF	1.21 [0.880]	1.14 [0.822]	1.16 [0.941]	0.99 [0.450]
RMIDAS-HF	0.92 [0.065]	0.85 [0.007]	0.96 [0.281]	0.90 [0.023]
<i>Panel C. VAR(3) dynamics</i>				
BU-Recursive	0.66 [0.000]	0.61 [0.000]	0.73 [0.000]	0.64 [0.000]
BUMIDAS	0.66 [0.000]	0.62 [0.000]	0.73 [0.000]	0.66 [0.000]
LF VAR	0.94 [0.011]	0.91 [0.032]	0.95 [0.081]	0.93 [0.023]
UMIDAS	1.08 [0.902]	1.06 [0.781]	1.05 [0.860]	1.01 [0.562]
RMIDAS	0.92 [0.002]	0.88 [0.011]	0.94 [0.024]	0.91 [0.004]
BUMIDAS-X	0.66 [0.000]	0.61 [0.000]	0.72 [0.000]	0.65 [0.000]
UMIDAS-HF	1.06 [0.762]	0.99 [0.452]	0.97 [0.371]	1.07 [0.820]
RMIDAS-HF	0.79 [0.000]	0.71 [0.000]	0.79 [0.000]	0.75 [0.000]

Notes: Entries report MSFE ratios relative to the monthly-average no-change forecast, with Diebold–Mariano p -values in brackets. BUMIDAS-X excludes oil; UMIDAS-HF and RMIDAS-HF use high-frequency FX. CADNEER and NOKNEER denote nominal effective exchange rates.

Table A16 isolates the role of temporal aggregation. Panel A evaluates the same baseline monthly-average forecasts relative to the end-of-month no-change forecast. Panel B repeats the exercise for end-of-period exchange rates; in this case, the low-frequency exchange-rate lags in

the VAR, UMIDAS, and RMIDAS specifications are also end-of-month observations rather than monthly averages. Consistent with the exchange-rate forecasting literature, none of the models significantly outperform the end-of-month no-change benchmark in either case. The reported p -values compare each forecast with BUMIDAS-BIC.

With end-of-period sampling, the low-frequency exchange-rate lag supplies the latest high-frequency target observation, largely eliminating the LF VAR and RMIDAS accuracy gaps relative to BUMIDAS-BIC. BUMIDAS-BIC remains more accurate than UMIDAS under both sampling schemes, consistent with the estimation cost of including extra unrestricted daily coefficients even when the latest target observation is available.

Table A16. Period-Average and End-of-Period Exchange Rate Forecasts vs. End-of-Month No-Change

Method	CAD	NOK	CADNEER	NOKNEER
<i>Panel A. Monthly-average exchange-rate target</i>				
RWWD	1.00	1.00	1.01	1.00
BU-Recursive	1.00	1.00	1.02	1.00
BUMIDAS-BIC	1.00	1.01	1.00	1.02
LF VAR	1.41 [0.000]	1.52 [0.000]	1.31 [0.000]	1.49 [0.000]
UMIDAS	1.46 [0.000]	1.63 [0.000]	1.37 [0.000]	1.60 [0.000]
RMIDAS	1.40 [0.000]	1.49 [0.000]	1.30 [0.000]	1.47 [0.000]
UMIDAS-HF	1.17 [0.001]	1.23 [0.012]	1.09 [0.148]	1.42 [0.001]
RMIDAS-HF	1.20 [0.008]	1.18 [0.024]	1.09 [0.052]	1.17 [0.021]
<i>Panel B. End-of-period exchange-rate target</i>				
RWWD	1.00	1.00	1.02	1.01
BU-Recursive	1.00	1.00	1.02	1.01
BUMIDAS-BIC	1.01	1.01	1.00	1.01
LF VAR	1.01 [0.541]	1.02 [0.132]	1.00 [0.738]	1.01 [0.424]
UMIDAS	1.07 [0.018]	1.09 [0.062]	1.07 [0.001]	1.09 [0.005]
RMIDAS	1.01 [0.355]	1.03 [0.087]	1.00 [0.710]	1.00 [0.653]
UMIDAS-HF	1.20 [0.000]	1.20 [0.012]	1.18 [0.001]	1.28 [0.002]
RMIDAS-HF	1.02 [0.206]	1.06 [0.029]	1.01 [0.345]	1.02 [0.199]

Notes: MSFE ratios are relative to the end-of-month no-change forecast. Brackets report DM p -values against BUMIDAS-BIC. Panel A forecasts monthly averages; Panel B forecasts end-of-period exchange rates. NEER denotes nominal effective exchange rates. UMIDAS-HF and RMIDAS-HF use high-frequency FX; their baseline counterparts use low-frequency FX.

A5.4 Inflation forecasts

The Bayesian LASSO uses the hierarchical shrinkage formulation of Park and Casella (2008), with a unit-scale half-Cauchy prior on global shrinkage, unit-scale Rayleigh priors on local scales, and a Jeffreys prior for the disturbance variance. The continuous prior shrinks coefficients toward zero without excluding them exactly.

Post-LASSO-BIC selects the LASSO penalty by minimizing BIC computed after refitting each selected specification by OLS. Forecasts use the corresponding post-LASSO estimates (Tibshirani, 1996; Zhang et al., 2010; Belloni and Chernozhukov, 2013).

BMIDAS-AGL follows Mogliani and Simoni (2021), using cubic Almon polynomials with the weight and its first derivative restricted to zero at the oldest lag. The resulting two coefficients per predictor form one group under an adaptive Bayesian group-LASSO normal-gamma prior. Lag windows contain 20 daily observations for oil prices, interest rates, and the BPP proxy, and three weekly observations for gasoline prices. Two inflation lags remain unpenalized. The disturbance variance has an inverse-gamma prior with shape and scale parameters of 10^{-3} , and group-specific shrinkage is estimated by stochastic approximation with step-size exponent 0.8. Target centering and regressor standardization use only the current estimation sample.

The UCM adopts the unobserved-components framework of Stock and Watson (2007), using a local-level model with a random-walk trend and independent Gaussian trend and transitory disturbances. Their variances are constant within each estimation window and reestimated by maximum likelihood at each forecast origin. Forecasts use the Kalman-filter estimate of the current trend.

The BVAR-CCM includes monthly CPI inflation, oil and gasoline price growth, and changes in the three-month Treasury bill rate, with 12 lags and a constant. In the spirit of Carriero et al. (2015), its conjugate normal-inverse-Wishart Minnesota prior has zero coefficient means, overall tightness

0.3162, lag-decay parameter 1, and constant-term tightness 100. The innovation covariance prior has six degrees of freedom and a diagonal scale matrix based on residual variances from separate AR(12) models.

Bayesian LASSO and BVAR-CCM estimation retain 10,000 posterior draws after 2,500 burn-in iterations. BMIDAS-AGL retains 40,000 draws by keeping every tenth iteration after 200,000 burn-in iterations. Bayesian forecasts are posterior predictive means.

A5.5 GDP Nowcasting

We construct three nowcasts of quarterly U.S. real GDP growth using the three monthly real-time vintages within each quarter. Let $\bar{y}_t = 100\Delta \log GDP_t$ denote quarterly real GDP growth and let $\mathbf{x}_{t,i}$ denote the vector of monthly proxies available at forecast origin $i = 1, 2, 3$ within quarter t . The vector contains industrial production, housing starts, nonfarm payroll employment, and capacity utilization.

The bridge models first complete the monthly indicator paths through the end of the target quarter. At each monthly vintage, each indicator x_s , where s indexes calendar months, is transformed to three-month growth, $g_s = 100(\log x_s - \log x_{s-3})$. An AR(p) with a constant is selected by BIC over $p = 1, \dots, 12$ using a common estimation sample. The selected model is then reestimated using all available observations and used to forecast recursively from the last observed month through the end of the quarter. Quarterly bridge predictors are the arithmetic averages of the three observed or forecast monthly growth rates.

The combined bridge equation regresses quarterly GDP growth on a constant, one lag of GDP growth, and the four contemporaneous quarterly bridge predictors. It is estimated by OLS using the GDP and predictor histories available in each vintage. The pooled bridge estimates the corresponding equation separately for each indicator and equally weights the four resulting GDP

nowcasts. In both versions, any unavailable lagged GDP growth is forecast recursively from the last observed quarter through the target quarter.

The MIDAS predictors use the month-over-month growth transformation. The qualitative findings are unchanged when both bridge and MIDAS models use either one-month or three-month growth transformations for the monthly indicators. At each forecast origin, the MIDAS specifications use the six most recent monthly observations from the current vintage, $\mathbf{x}_{t,i}, L\mathbf{x}_{t,i}, \dots, L^5\mathbf{x}_{t,i}$, with L defined in Section 2. Following Clements and Galvão (2008), Foroni et al. (2015a), and Schumacher (2016), we consider combined and pooled versions of BUMIDAS-BIC, UMIDAS, and RMIDAS. The combined specifications take the general form

$$\bar{y}_t = \alpha + \rho\bar{y}_{t-1} + B(L)\mathbf{x}_{t,i} + u_t, \quad (48)$$

where the parameterization of the high-frequency lag polynomial $B(L)$ differs across methods. The pooled specifications estimate the corresponding model separately for each element of $\mathbf{x}_{t,i}$ and equally weight the four resulting forecasts,

$$\hat{y}_{t,i}^P = \frac{1}{4} \sum_{r=1}^4 \hat{y}_{r,t,i}. \quad (49)$$

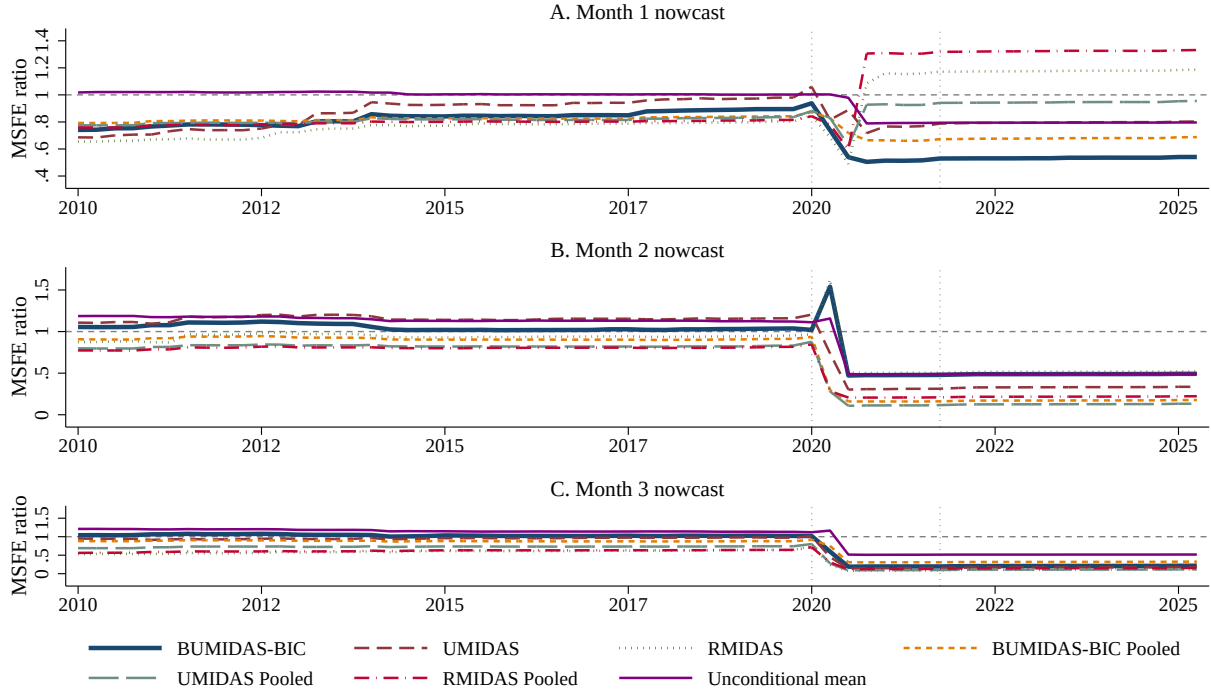
BUMIDAS-BIC always includes the most recent observation of each predictor in $\mathbf{x}_{t,i}$. Earlier monthly observations, $L\mathbf{x}_{t,i}, \dots, L^5\mathbf{x}_{t,i}$, are added sequentially within each predictor and retained while they lower the BIC; the search stops at the first additional lag that fails to improve the criterion. The selected specification is then reestimated by OLS. For the pooled forecasts, selection is conducted separately for each predictor before the four forecasts are combined according to (49).

UMIDAS follows Foroni et al. (2015a) and includes all six monthly observations with unrestricted coefficients. RMIDAS uses the same six-month history but restricts the coefficients for

each predictor using a quadratic Almon polynomial.

All models are reestimated with an expanding window using only the GDP and predictor vintages available at forecast origin. Forecast accuracy is evaluated against the 2026m3 vintage of the quarterly GDP growth.

Figure A3. Evolution of Real-Time GDP Nowcast Accuracy



Notes: Expanding MSFE ratios relative to the AR(1) benchmark for nowcasts formed in each month of the quarter. Paths are displayed from 2010Q1 to limit sensitivity to initial observations. Values below one indicate lower MSFE than the AR(1); vertical lines delimit 2020Q1–2021Q4.

Unlike the interest-rate and exchange-rate results, Figure A3 shows that COVID-19 materially changes the forecast ranking. The expanding MSFE ratio of the recursively estimated unconditional-mean forecast falls below one during the pandemic and remains there afterward, consistent with its lower sensitivity to the large but transitory contraction and rebound. At the first forecast origin, BUMIDAS-BIC becomes the most accurate specification during COVID-19 and retains the lowest MSFE ratio through the end of the sample.

Table A17 compares the AR(1) baseline used in Table 9 and Figure A3 with alternative low-

frequency benchmarks. The AR(1) outperforms the AR(2) and BIC-selected AR specification at all three nowcast origin months. The recursively estimated unconditional mean of quarterly GDP growth yields a lower MSFE than the AR(1), although the difference is not statistically significant. We retain the AR(1) for consistency with the real-time GDP forecasting exercises of Clements and Galvão (2008, 2009) and the AR benchmark used by Schumacher (2016).

Table A17. Alternative Low-Frequency GDP Benchmarks

	AR(1)	AR(2)	AR-BIC	No Change	Unconditional mean
Month 1	1.00	1.08 (0.873)	1.03 (0.876)	1.65 (0.879)	0.80 (0.153)
Month 2	1.00	1.06 (0.861)	1.01 (0.959)	1.07 (0.598)	0.50 (0.162)
Month 3	1.00	1.07 (0.862)	1.01 (0.955)	1.12 (0.675)	0.52 (0.163)

Notes: Entries report MSFE ratios relative to the recursively estimated AR(1). Parentheses report one-sided Diebold–Mariano p -values for equal predictive accuracy relative to the AR(1). AR-BIC selects among autoregressive specifications by BIC at each forecast origin.